Prepared for

Community Foundation for Muskegon County

Serving the Community Foundations of Muskegon, Mason, and Oceana Counties

April 25, 2019



Vanguard

Advice services offered through Vanguard Institutional Advisory Services® are provided by Vanguard Advisers, Inc., a registered investment advisor.

For institutional use only. Not for distribution to retail investors.

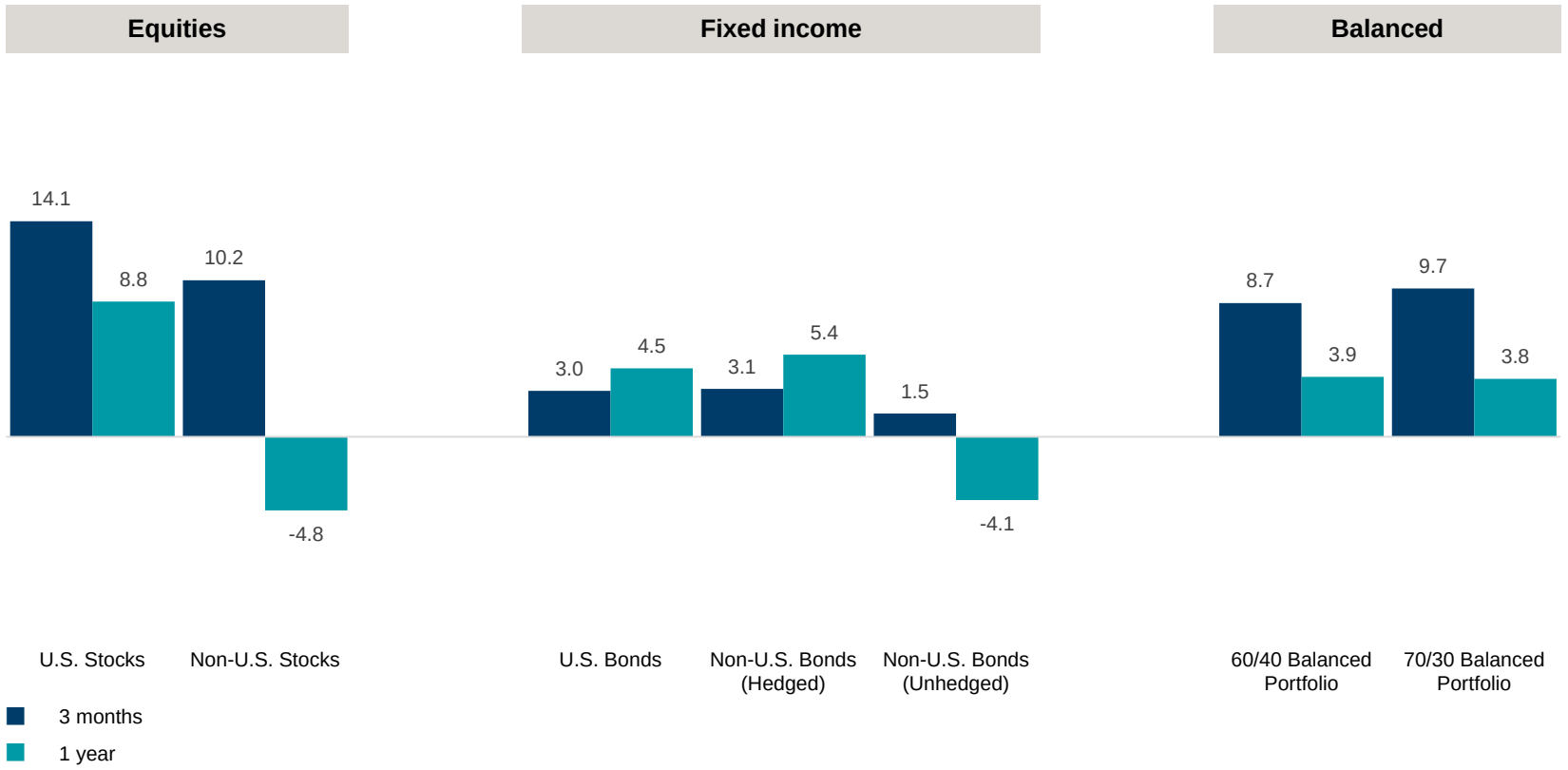
Agenda

I. Market recap
II. Portfolio performance and asset allocation
III. Market and economic review
IV. Fund summaries

Market recap

Equities and fixed income continue to have a good start in 2019

Global market returns as of March 31, 2019 (%)

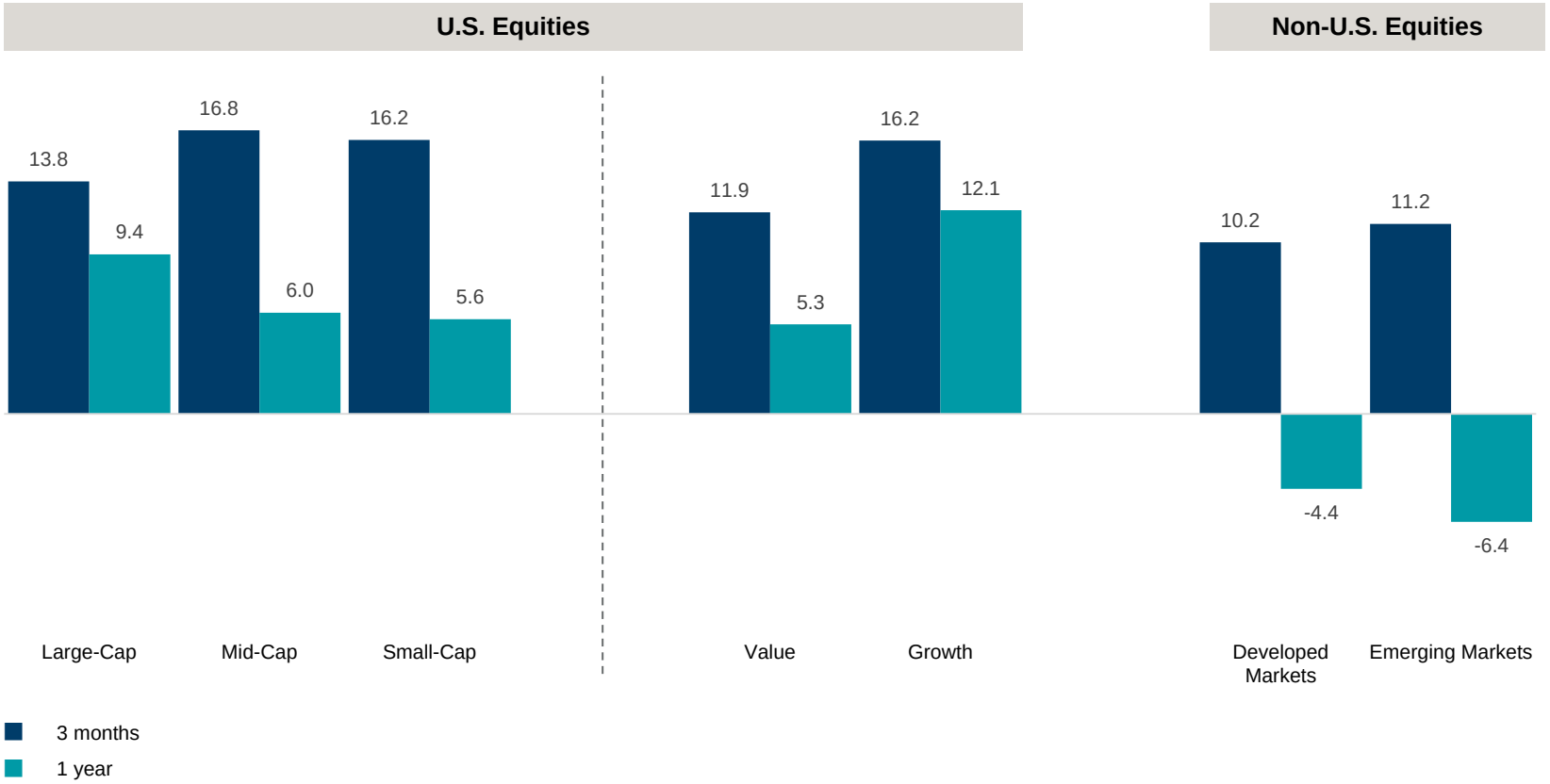


Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: Bloomberg Barclays, FTSE, MSCI, Russell, CRSP and Dow Jones.
US Stocks (CRSP US Total Market Index), Non-US Stocks (FTSE Global All-Cap ex-US Index), US Bonds (Bloomberg Barclays US Aggregate Float Adjusted Index), Non-US Bonds hedged (Bloomberg Barclays Global Aggregate ex-USD Float Adjusted RIC Capped Index hedged), Non-US Bonds unhedged (Bloomberg Barclays Global Aggregate Index ex-USD).
* 60/40 balanced portfolio Static Composite (36% U.S. stocks, 24% International stocks, and 28% Investment-grade U.S. bonds, 12% Investment-grade international bonds).
** 70/30 balanced portfolio Static Composite (42% U.S. stocks, 28% International stocks, and 21% Investment-grade U.S. bonds, 9% Investment-grade international bonds).

All segments of U.S. and Non-U.S. equities had double digit returns over the past 3 months, however, Non-U.S have lagged over the past year

Global equity market returns as of March 31, 2019 (%)

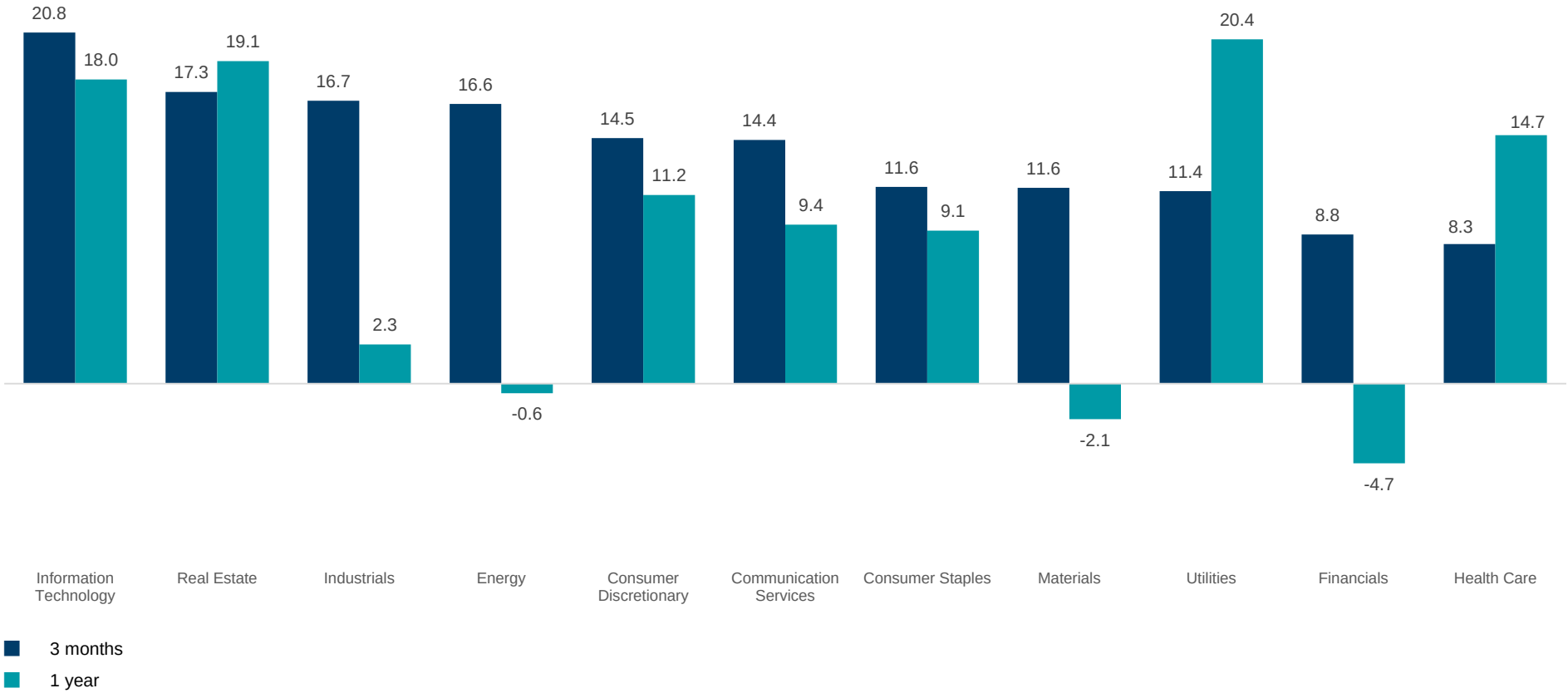


Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: FTSE, MSCI, Russell, CRSP and Dow Jones.
Large-cap (CRSP US Large Cap Index), Mid-cap (CRSP US Mid Cap Index), Small-cap (CRSP US Small Cap Index); Value (Russell 3000 Value Index), Growth (Russell 3000 Growth Index); Developed markets (FTSE Developed All Cap ex-US Index), Emerging markets (FTSE Emerging Markets All Cap China A Inclusion Index).

Within the U.S. equity market, over the last 3 months information technology and real estate have outperformed, while over the past 12 months utilities have led performance

U.S. equity sector returns as of March 31, 2019 (%)

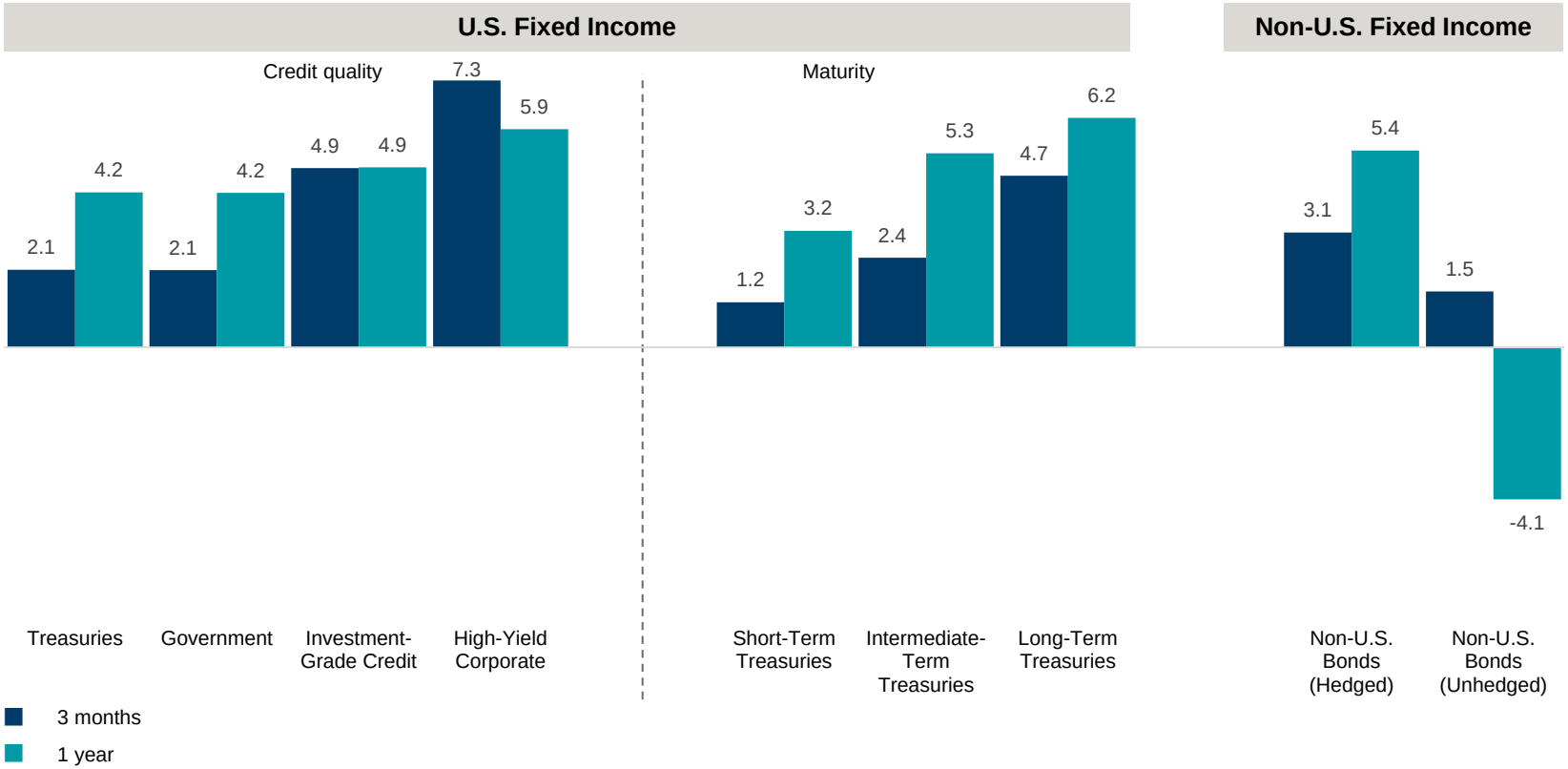


Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: FactSet and Vanguard.
U.S. markets measured by CRSP US Total Market Index. Past performance is no guarantee of future returns.

All segments of the fixed income markets except (Unhedged) Non- U.S. bonds had positive returns the past 3 and 12 months as fears of rising rates have lowered

Global fixed income market returns as of March 31, 2019 (%)

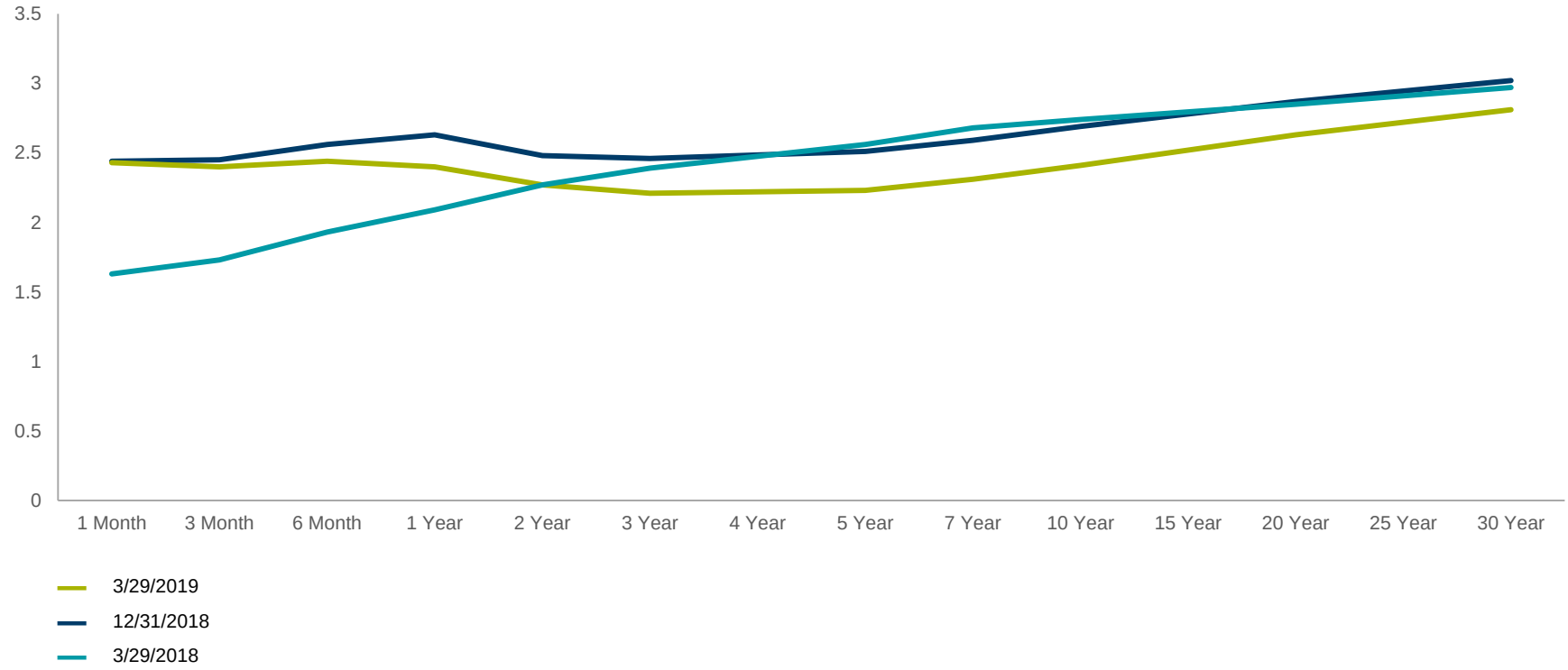


Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: Bloomberg Barclays.
Treasuries-Government-Investment Grade Corporates-High Yield (Bloomberg Barclays US Treasury/Government/Credit/Corporate High Yield Indices); Short-Inter-Long-term Treasuries (Bloomberg Barclays US 1-5/5-10/Long Year Treasury Indices)
Non-US Bonds hedged (Bloomberg Barclays Global Aggregate ex-USD Float Adjusted RIC Capped Index hedged), Non-US Bonds unhedged (Bloomberg Barclays Global Aggregate Index ex-USD).

The U.S. Treasury yield curve continued to flatten

Yield (%) and change (bps)	1 month	3 month	6 month	1 year	2 year	3 year	5 year	7 year	10 year	20 year	30 year
Current yield (%)	**2.43**	**2.4**	**2.44**	**2.4**	**2.27**	**2.21**	**2.23**	**2.31**	**2.41**	**2.63**	**2.81**
3 month change	-1	-5	-12	-23	-21	-25	-28	-28	-28	-24	-21
12 month change	80	67	51	31	0	-18	-33	-37	-33	-22	-16



Source: U.S. Treasury.

Portfolio performance and asset allocation

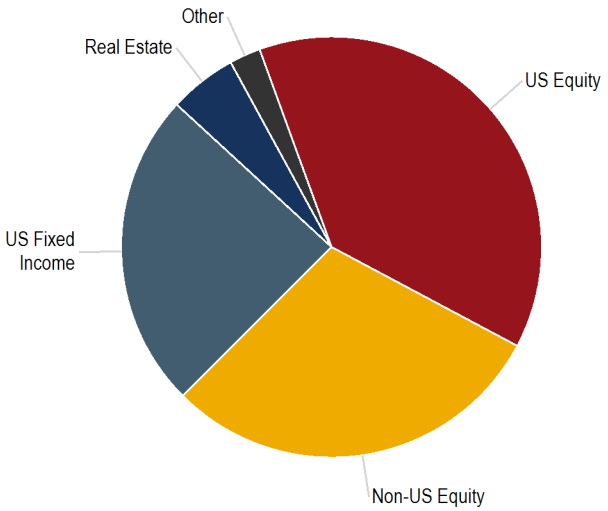
Total Portfolio Performance & Asset Allocation

Performance Summary ending March 31, 2019

	Market Value ($)	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**180,361,639**	**1.47**	**9.82**	**4.11**	**8.07**	**--**	**--**	**6.17**	**Apr-14**
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY (Net)		**1.47**	**9.81**	**4.06**	**8.02**	**--**	**--**	**6.12**	
Composite Benchmark		1.44	10.23	4.43	7.90	--	--	6.10	Apr-14

- Composite Benchmark = 39% Spliced Total Stock Market Index / 31% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 5% Real Estate Spliced Index

Current Allocation as of March 31, 2019



	Current $	Current %	Policy	Difference*
US Equity	$69,168,238	38.3%	39.0%	-0.7%
Non-US Equity	$53,571,228	29.7%	31.0%	-1.3%
US Fixed Income	$43,875,671	24.3%	22.6%	1.7%
Real Estate	$9,446,502	5.2%	5.0%	0.2%
Other	$4,300,000	2.4%	2.4%	0.0%
Total	**$180,361,639**	**100.0%**	**100.0%**	

*Difference between Policy and Current Allocation
** “Other” represents the foundation’s private loans, which are not being managed by Vanguard.

Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
Net of Fee returns reflect the deduction of fund expense ratios, any purchase or redemption fees, and VIAS advisory fee applied to the client portfolio.
Returns greater than one year represent annualized returns. Returns less than one year represent cumulative returns.

Performance Summary (Gross of Advisory Fees) ending March 31, 2019

	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**180,361,639**	**100.00**	**1.47**	**9.82**	**4.11**	**8.07**	**--**	**--**	**6.17**	**Apr-14**
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY (Net)			**1.47**	**9.81**	**4.06**	**8.02**	**--**	**--**	**6.12**	
Composite Benchmark			1.44	10.23	4.43	7.90	--	--	6.10	Apr-14
MUSKEGON COUNTY SPLICED TOTAL			1.26	8.14	1.64	7.04	--	--	4.26	Apr-14
Total Equity	**122,739,466**	**68.05**	**1.13**	**12.36**	**2.61**	**11.20**	**--**	**--**	**7.26**	**Apr-14**
Equity Domestic	**69,168,238**	**38.35**	**1.45**	**14.04**	**8.81**	**13.53**	**--**	**--**	**10.52**	**Apr-14**
Spliced Total Stock Market Index			1.45	14.06	8.82	13.53	10.35	16.07	10.52	Apr-14
Equity International	**53,571,228**	**29.70**	**0.73**	**10.26**	**-5.18**	**8.08**	**--**	**--**	**2.58**	**Apr-14**
Spliced Total International Stock Index			0.57	10.21	-4.81	7.99	2.84	8.96	2.65	Apr-14
Total Fixed Income	**43,875,671**	**24.33**	**1.97**	**2.95**	**4.47**	**1.99**	**--**	**--**	**2.54**	**Apr-14**
Fixed Income Domestic	**43,875,671**	**24.33**	**1.97**	**2.95**	**4.47**	**1.99**	**--**	**--**	**2.54**	**Apr-14**
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			1.95	3.00	4.46	2.05	2.73	3.79	2.60	Apr-14
Total Real Estate	**9,446,502**	**5.24**	**4.16**	**17.28**	**20.01**	**5.73**	**--**	**--**	**8.22**	**Apr-14**
Real Estate Domestic	**9,446,502**	**5.24**	**4.16**	**17.28**	**20.01**	**5.73**	**--**	**--**	**8.22**	**Apr-14**
Real Estate Spliced Index			4.19	17.28	20.08	5.81	8.89	18.36	8.32	Apr-14
Private Loans	**4,300,000**	**2.38**	**0.27**	**0.81**	**3.29**	**--**	**--**	**--**	**3.29**	**Mar-18**

Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
Net of Fee returns reflect the deduction of fund expense ratios, any purchase or redemption fees, and VIAS advisory fee applied to the client portfolio.
Returns greater than one year represent annualized returns. Returns less than one year represent cumulative returns.

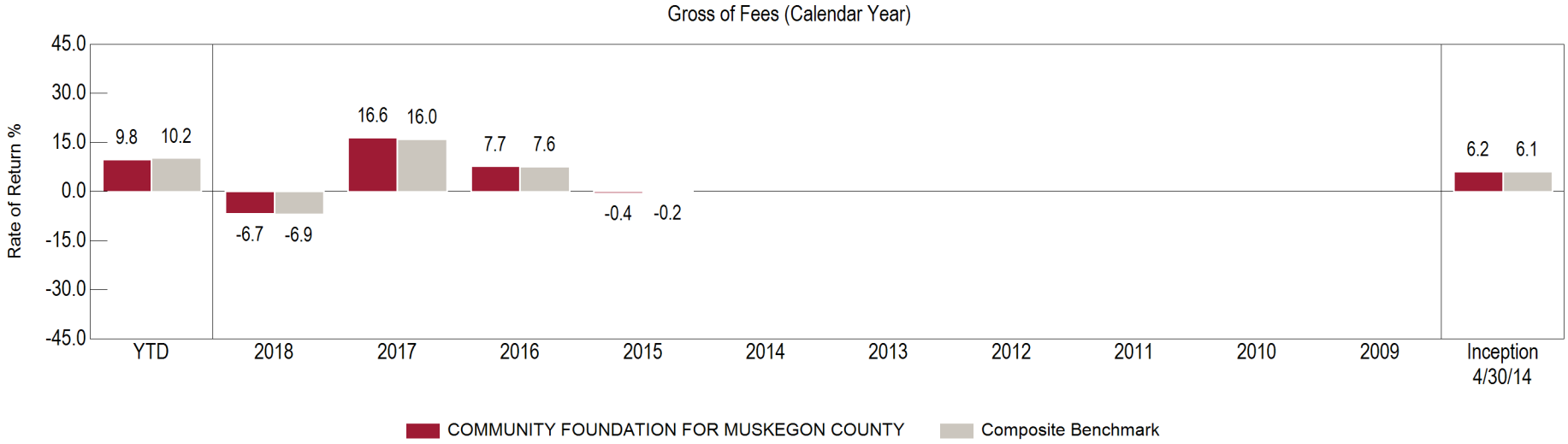
Performance Summary (Gross of Advisory Fees) ending March 31, 2019

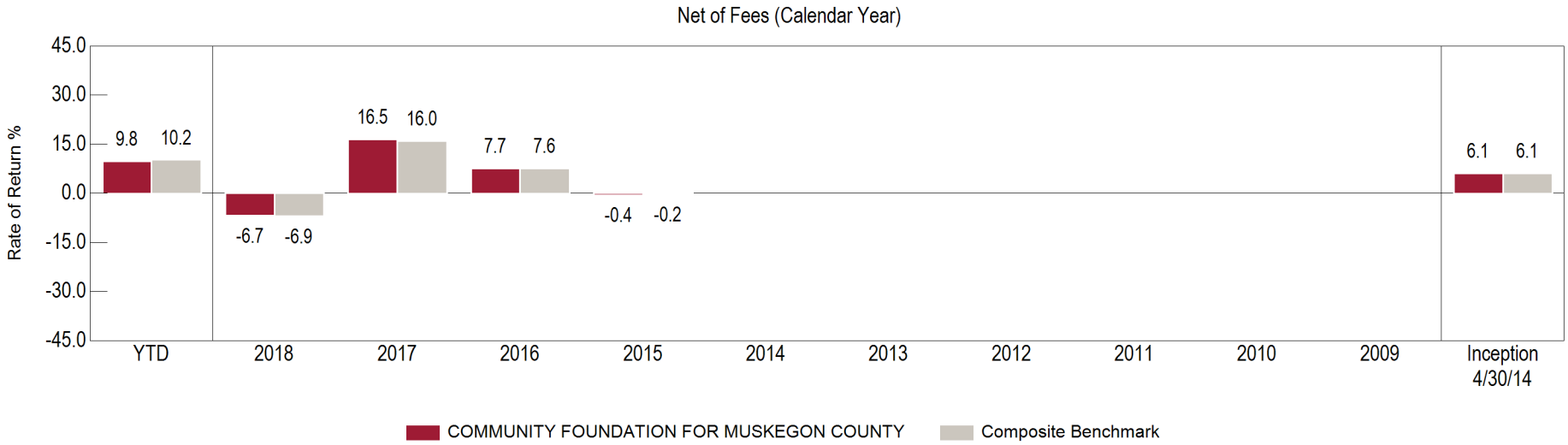
	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**180,361,639**	**100.00**	**1.47**	**9.82**	**4.11**	**8.07**	**--**	**--**	**6.17**	**Apr-14**
Composite Benchmark			1.44	10.23	4.43	7.90	--	--	6.10	Apr-14
MUSKEGON COUNTY SPLICED TOTAL			1.26	8.14	1.64	7.04	--	--	4.26	Apr-14
Total Equity	**122,739,466**	**68.05**	**1.13**	**12.36**	**2.61**	**11.20**	**--**	**--**	**7.26**	**Apr-14**
Equity Domestic	**69,168,238**	**38.35**	**1.45**	**14.04**	**8.81**	**13.53**	**--**	**--**	**10.52**	**Apr-14**
Spliced Total Stock Market Index			1.45	14.06	8.82	13.53	10.35	16.07	10.52	Apr-14
Vanguard® Total Stock Market Index Fund Institutional Shares	69,168,238	38.35	1.45	14.04	8.81	13.52	10.34	16.06	10.51	Apr-14
Spliced Total Stock Market Index			1.45	14.06	8.82	13.53	10.35	16.07	10.52	Apr-14
Multi-Cap Core Funds Average			0.78	13.09	4.95	10.82	7.52	13.66	7.72	Apr-14
Equity International	**53,571,228**	**29.70**	**0.73**	**10.26**	**-5.18**	**8.08**	**--**	**--**	**2.58**	**Apr-14**
Spliced Total International Stock Index			0.57	10.21	-4.81	7.99	2.84	8.96	2.65	Apr-14
Vanguard® Total International Stock Index Fund Inst Shares	53,571,228	29.70	0.73	10.26	-5.18	8.08	2.79	8.93	2.57	Apr-14
Spliced Total International Stock Index			0.57	10.21	-4.81	7.99	2.84	8.96	2.65	Apr-14
International Funds Average			0.82	10.85	-5.84	6.64	2.16	8.64	2.04	Apr-14
Total Fixed Income	**43,875,671**	**24.33**	**1.97**	**2.95**	**4.47**	**1.99**	**--**	**--**	**2.54**	**Apr-14**
Fixed Income Domestic	**43,875,671**	**24.33**	**1.97**	**2.95**	**4.47**	**1.99**	**--**	**--**	**2.54**	**Apr-14**
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			1.95	3.00	4.46	2.05	2.73	3.79	2.60	Apr-14
Vanguard® Total Bond Market Index Fund Institutional Shares	43,875,671	24.33	1.97	2.95	4.47	2.00	2.68	3.71	2.56	Apr-14
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			1.95	3.00	4.46	2.05	2.73	3.79	2.60	Apr-14
Spliced Intermediate-Term Investment-Grade Debt Funds Average			1.81	3.18	3.96	2.11	2.43	4.50	2.30	Apr-14
Total Real Estate	**9,446,502**	**5.24**	**4.16**	**17.28**	**20.01**	**5.73**	**--**	**--**	**8.22**	**Apr-14**
Real Estate Domestic	**9,446,502**	**5.24**	**4.16**	**17.28**	**20.01**	**5.73**	**--**	**--**	**8.22**	**Apr-14**
Real Estate Spliced Index			4.19	17.28	20.08	5.81	8.89	18.36	8.32	Apr-14
Vanguard® Real Estate Index Fund Institutional Shares	9,446,502	5.24	4.16	17.28	20.01	5.74	8.80	18.35	8.23	Apr-14
Real Estate Spliced Index			4.19	17.28	20.08	5.81	8.89	18.36	8.32	Apr-14
Real Estate Funds Average			3.45	15.70	16.23	5.49	7.98	17.26	7.48	Apr-14

Performance Summary (Gross of Advisory Fees) ending March 31, 2019

	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
Private Loans	**4,300,000**	**2.38**	**0.27**	**0.81**	**3.29**	**--**	**--**	**--**	**3.29**	**Mar-18**
Private Loan 1	2,300,000	1.28	0.27	0.81	3.29	--	--	--	3.29	Mar-18
Private Loan 2	2,000,000	1.11	0.27	0.81	3.29	--	--	--	3.29	Mar-18

Total Portfolio Performance





Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
Net of Fee returns reflect the deduction of fund expense ratios, any purchase or redemption fees, and VIAS advisory fee applied to the client portfolio.
Returns greater than one year represent annualized returns. Returns less than one year represent cumulative returns.

Cash Flow Summary

	Last Month	Last Three Months	Year-To-Date	One Year
Beginning Market Value	$178,089,967	$164,471,143	$164,471,143	$173,280,974
Net Cash Flow	-$338,354	-$319,489	-$319,489	-$324,129
Capital Appreciation	$1,884,335	$15,250,672	$15,250,672	$2,813,104
Income	$725,692	$959,313	$959,313	$4,591,690
Ending Market Value	$180,361,639	$180,361,639	$180,361,639	$180,361,639

Month Ending March 31, 2019

	Beginning Market Value	Net Cash Flow	Capital Appreciation	Income	Ending Market Value
Private Loan 1	$2,300,000	-$6,229	$0	$6,229	$2,300,000
Private Loan 2	$2,000,000	-$5,417	$0	$5,417	$2,000,000
Vanguard® Real Estate Index Fund Institutional Shares	$9,087,046	-$18,559	$311,020	$66,996	$9,446,502
Vanguard® Total Bond Market Index Fund Institutional Shares	$43,357,276	-$331,441	$741,202	$108,635	$43,875,671
Vanguard® Total International Stock Index Fund Inst Shares	$53,180,573	$7,095	$211,894	$171,666	$53,571,228
Vanguard® Total Stock Market Index Fund Institutional Shares	$68,165,073	$16,196	$620,220	$366,750	$69,168,238
Total	**$178,089,967**	**-$338,354**	**$1,884,335**	**$725,692**	**$180,361,639**

Investment Expense Analysis as of March 31, 2019

Name	Market Value	% of Portfolio	Expense Ratio
Total Equity	**$122,739,466**	**68.1%**	
Equity Domestic	**$69,168,238**	**38.3%**	
Vanguard® Total Stock Market Index Fund Institutional Shares	$69,168,238	38.3%	0.035%
Equity International	**$53,571,228**	**29.7%**	
Vanguard® Total International Stock Index Fund Inst Shares	$53,571,228	29.7%	0.080%
Total Fixed Income	**$43,875,671**	**24.3%**	
Fixed Income Domestic	**$43,875,671**	**24.3%**	
Vanguard® Total Bond Market Index Fund Institutional Shares	$43,875,671	24.3%	0.040%
Total Real Estate	**$9,446,502**	**5.2%**	
Real Estate Domestic	**$9,446,502**	**5.2%**	
Vanguard® Real Estate Index Fund Institutional Shares	$9,446,502	5.2%	0.100%
Private Loans	**$4,300,000**	**2.4%**	
Private Loan 1	$2,300,000	1.3%	
Private Loan 2	$2,000,000	1.1%	
Total	**$180,361,639**	**100.0%**	**0.052%**

Market Performance as of March 31, 2019

Name	Mar-19	Last 3 Months	1 Yr	3 Yrs	5 Yrs	10 Yrs
US Equity						
CRSP US Total Market TR USD	1.4	14.1	8.8	13.5	10.3	16.0
S&P 500	1.9	13.6	9.5	13.5	10.9	15.9
S&P 400 MidCap	-0.6	14.5	2.6	11.2	8.3	16.3
S&P 600 SmallCap	-3.3	11.6	1.6	12.6	8.5	17.0
International Equity						
MSCI Emerging Markets	0.8	9.9	-7.4	10.7	3.7	8.9
MSCI Emerging Markets NR LCL	1.4	10.1	-1.7	11.3	7.2	10.2
MSCI EAFE	0.6	10.0	-3.7	7.3	2.3	9.0
MSCI EAFE NR LCL	1.3	10.6	2.8	8.5	6.0	9.8
MSCI ACWI ex USA	0.6	10.3	-4.2	8.1	2.6	8.8
Fixed Income Domestic						
BBgBarc US Aggregate TR	1.9	2.9	4.5	2.0	2.7	3.8
BBgBarc US Corporate 1-5 Years TR	1.1	2.6	4.4	2.4	2.3	4.3
BBgBarc US Credit/Corp 5-10 Yr TR	2.3	5.4	6.1	3.5	4.0	7.3
BBgBarc US Corporate Long TR	4.3	8.0	4.4	5.3	5.3	9.2
BBgBarc US Govt/Credit Long TR	4.7	6.5	5.2	3.8	5.3	7.2
BBgBarc US Treasury Strips 20-30 Yr Equal Parity TR	7.5	6.1	6.5	1.7	7.7	6.3
BBgBarc US High Yield TR	0.9	7.3	5.9	8.6	4.7	11.3
BBgBarc US Govt TR	1.9	2.1	4.2	1.1	2.1	2.4
BBgBarc US Credit TR	2.4	4.9	4.9	3.5	3.6	6.2
BBgBarc US Treasury 1-5 Yr TR	0.9	1.2	3.2	1.0	1.3	1.4
BBgBarc US Treasury 5-10 Yr TR	2.1	2.4	5.3	1.0	2.7	3.1
BBgBarc US Treasury Long TR	5.3	4.7	6.2	1.5	5.4	5.1
BBgBarc US Treasury TIPS 0-5 Yr TR	0.8	1.7	2.0	1.4	0.9	1.9
BBgBarc US TIPS TR	1.8	3.2	2.7	1.7	1.9	3.4
Fixed Income International						
BBgBarc Global Aggregate ex US Tres Hedged TR	1.7	3.0	5.2	3.3	4.3	4.3
BBgBarc Emerging Markets TR	1.4	5.4	4.4	5.4	4.8	8.5
REIT						
MSCI US REIT Gross	3.3	16.3	20.7	6.0	9.0	18.5

Benchmark History as of March 31, 2019

COMMUNITY FOUNDATION FOR MUSKEGON COUNTY		
8/1/2018	Present	39% Spliced Total Stock Market Index / 31% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 5% Real Estate Spliced Index
7/1/2018	7/31/2018	39% Spliced Total Stock Market Index / 30% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 6% Real Estate Spliced Index
6/1/2018	6/30/2018	39% Spliced Total Stock Market Index / 29% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 7% Real Estate Spliced Index
4/1/2018	5/31/2018	39% Spliced Total Stock Market Index / 28% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 8% Real Estate Spliced Index
3/1/2018	3/31/2018	39% Spliced Total Stock Market Index / 27% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 9% Real Estate Spliced Index
9/1/2017	2/28/2018	39% Spliced Total Stock Market Index / 26% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
8/1/2017	8/31/2017	38% Spliced Total Stock Market Index / 25.33% Spliced Total International Stock Index / 26.67% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
7/1/2017	7/31/2017	37.5% Spliced Total Stock Market Index / 25% Spliced Total International Stock Index / 27.5% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
5/1/2014	6/30/2017	10% REIT Spliced Index / 30% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 24% Spliced Total International Stock Index / 36% Spliced Total Stock Market Index
Total Equity		
		N/A
Equity Domestic		
4/30/2014	Present	100% CRSP US Total Market TR USD
Vanguard® Total Stock Market Index Fund Institutional Shares		
4/30/2014	Present	100% CRSP US Total Market TR USD
Equity International		
4/30/2014	Present	100% FTSE Global All-Cap ex-US Index
Vanguard® Total International Stock Index Fund Inst Shares		
4/30/2014	Present	100% FTSE Global All-Cap ex-US Index

Benchmark History as of March 31, 2019

Total Fixed Income		
		N/A
Fixed Income Domestic		
4/30/2014	Present	100% BBgBarc US Aggregate Float Adjusted TR
Vanguard® Total Bond Market Index Fund Institutional Shares		
4/30/2014	Present	100% BBgBarc US Aggregate Float Adjusted TR
Total Real Estate		
		N/A
Real Estate Domestic		
7/25/2018	Present	100% MSCI US IM Real Estate 25/50 Index
1/26/2018	7/24/2018	100% MSCI US IM Real Estate 25/50 Tran Index
4/30/2014	1/25/2018	100% MSCI US REIT Gross
Vanguard® Real Estate Index Fund Institutional Shares		
7/25/2018	Present	100% MSCI US IM Real Estate 25/50 Index
1/26/2018	7/24/2018	100% MSCI US IM Real Estate 25/50 Tran Index
4/30/2014	1/25/2018	100% MSCI US REIT Gross
Private Loans		
		N/A
Private Loan 1		
		N/A
Private Loan 2		
		N/A

Asset allocation summary of CFMC VIAS composite portfolio
As of March 31, 2019

	Market value as of March 31, 2019	Policy target allocation	Actual weight	Variance (%)	Variance ($)
Domestic equity					
Total Stock Market Index Fund Institutional Shares	$69,168,238	39.0%	38.3%	–0.7%	–$1,172,801
Domestic equity subtotal	*$69,168,238*	*39.0%*	*38.3%*	*–0.7%*	
International equity					
Total International Stock Index Fund Institutional Shares	$53,571,228	31.0%	29.7%	–1.3%	–$2,340,880
International equity subtotal	*$53,571,228*	*31.0%*	*29.7%*	*–1.3%*	
Fixed income					
Total Bond Market Index Fund Institutional Shares	$43,875,671	22.6%	24.3%	1.7%	$3,113,941
Private loans*	$4,300,000	2.4%	2.4%	0.0%	–$28,679
Fixed income subtotal	*$48,175,671*	*25.0%*	*26.7%*	*1.7%*	
Other investments					
Real Estate Index Fund Institutional Shares	$9,446,502	5.0%	5.2%	0.2%	$428,420
Other investments subtotal	*$9,446,502*	*5.0%*	*5.2%*	*0.2%*	
Total portfolio	**$180,361,639**	**100.0%**	**100.0%**		

* Private loans are not managed by Vanguard.

Quarterly summary of combined historical performance as of March 31, 2019

	Annualized return						Calendar return								
	QTR	YTD	1 year	3 year	5 year	10 year	2018	2017	2016	2015	2014	2013	2012	2011	2010
CFMC Composite Portfolio	9.81	9.81	4.06	8.02	5.88	9.99	–6.71	16.50	7.67	–0.44	5.68	12.35	12.21	–3.47	13.95
*CFMC Composite Benchmark**	*10.23*	*10.23*	*4.43*	*7.90*	*6.10*	*10.83*	*–6.87*	*15.96*	*7.58*	*–0.22*	*7.08*	*19.26*	*12.92*	*0.70*	*12.98*

Notes:
Composite performance data consists of previous providers' portfolio and benchmark returns through April 2014; VIAS returns thereafter. Data prior to May 2014 is based upon information taken from the previous providers' quarterly performance reports and is provided as an estimate only. Composite does not include segregated accounts or pledge holdings.
Return calculations are derived from Morningstar Direct
* Composite Benchmark allocations over time:
50% S&P 500 / 10% Russell 2000 / 10% MSCI EAFE / 30% LB AGG from March 2004 to March 2005;
45% S&P 500 / 12.5% Russell 2000 / 12.5% MSCI EAFE / 30% LB AGG from March 2005 to December 2007;
55% Russell 3000 / 15% MSCI AC World Index ex-US / 30% LB AGG from December 2007 to December 2008;
55% Russell 3000 / 15% MSCI AC World Index ex-US / 30% Barclays AGG from December 2008 to April 2014;
36% Spliced Total Stock Market Index (CRSP US Total Market Index) / 30% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index) / 24% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 10% REIT Spliced Index (MSCI US REIT Index) from May 2014 to June 2017.
39% Spliced Total Stock Market Index (CRSP US Total Market Index) / 25% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index) / 26% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 10% REIT Spliced Index (MSCI US REIT Index) from July 2017 to February 2018.
39% Spliced Total Stock Market Index (CRSP US Total Market Index) / 25% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index) / 31% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 5% REIT Spliced Index (MSCI US REIT Index) from March 2018 to the present.

Legal

For more information about Vanguard funds or non-Vanguard funds offered through Vanguard Brokerage Services, visit vanguard.com or call your Investment Consultant or Relationship Manager to obtain a prospectus or, if available, a summary prospectus. Investment objectives, risks, charges, expenses, and other important information are contained in the prospectus; read and consider it carefully before investing. Total Portfolio Net of Fees returns reflect the deduction of fund expense ratios, purchase or redemption fees, and any advisory service fee applied to the client portfolio.

Total Portfolio returns represent client-specific time-weighted returns (TWR) are presented gross of any applicable service fees with the exception of mutual fund expense ratios and other security-level expenses.

Client performance inception date is generally the first month-end after initial funding. Mutual funds and all investments are subject to risk, including the possible loss of the money you invest. Diversification does not ensure a profit or protect against a loss.

Performance figures assume the reinvestment of dividends and capital gains distributions. The fund performance percentages are based on fund total return data, adjusted for expenses, obtained from Lipper, a Thomson Reuters Company. The total return data was not adjusted for fees and loads.

Benchmark comparative indexes represent unmanaged or average returns on various financial assets, which can be compared with funds' total returns for the purpose of measuring relative performance.

The index is a product of S&P Dow Jones Indices LLC (“SPDJI”), and has been licensed for use by Vanguard. Standard & Poor’s® and S&P® are registered trademarks of Standard & Poor’s Financial Services LLC (“S&P”); Dow Jones® is a registered trademark of Dow Jones Trademark Holdings LLC (“Dow Jones”); S&P® and S&P 500® are trademarks of S&P; and these trademarks have been licensed for use by SPDJI and sublicensed for certain purposes by Vanguard. Vanguard product(s) are not sponsored, endorsed, sold or promoted by SPDJI, Dow Jones, S&P, their respective affiliates and none of such parties make any representation regarding the advisability of investing in such product(s) nor do they have any liability for any errors, omissions, or interruptions of the index

The funds or securities referred to herein are not sponsored, endorsed, or promoted by MSCI, and MSCI bears no liability with respect to any such funds or securities. For any such funds or securities, the prospectus or the Statement of Additional Information contains a more detailed description of the limited relationship MSCI has with The Vanguard Group and any related funds.

The Russell Indexes and Russell® are registered trademarks of Russell Investments and have been licensed for use by The Vanguard Group. The products are not sponsored, endorsed, sold or promoted by Russell Investments and Russell Investments makes no representation regarding the advisability of investing in the products.

London Stock Exchange Group companies include FTSE International Limited ("FTSE"), Frank Russell Company ("Russell"), MTS Next Limited ("MTS"), and FTSE TMX Global Debt Capital Markets Inc. ("FTSE TMX"). All rights reserved. "FTSE®", "Russell®", "MTS®", "FTSE TMX®" and "FTSE Russell" and other service marks and trademarks related to the FTSE or Russell indexes are trademarks of the London Stock Exchange Group companies and are used by FTSE, MTS, FTSE TMX and Russell under license. All information is provided for information purposes only. No responsibility or liability can be accepted by the London Stock Exchange Group companies nor its licensors for any errors or for any loss from use of this publication. Neither the London Stock Exchange Group companies nor any of its licensors make any claim, prediction, warranty or representation whatsoever, expressly or impliedly, either as to the results to be obtained from the use of the FTSE Indexes or the fitness or suitability of the Indexes for any particular purpose to which they might be put.

BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. BARCLAYS® is a trademark and service mark of Barclays Bank Plc, used under license. Bloomberg Finance L.P. and its affiliates, including Bloomberg Index Services Limited ("BISL") (collectively, "Bloomberg"), or Bloomberg's licensors own all proprietary rights in the Bloomberg Barclays Indices.

The products are not sponsored, endorsed, issued, sold or promoted by "Bloomberg or Barclays". Bloomberg and Barclays make no representation or warranty, express or implied, to the owners or purchasers of the products or any member of the public regarding the advisability of investing in securities generally or in the products particularly or the ability of the Bloomberg Barclays Indices to track general bond market performance. Neither Bloomberg nor Barclays has passed on the legality or suitability of the products with respect to any person or entity. Bloomberg's only relationship to Vanguard and the products are the licensing of the Bloomberg Barclays Indices which are determined, composed and calculated by BISL without regard to Vanguard or the products or any owners or purchasers of the products. Bloomberg has no obligation to take the needs of the products into consideration in determining, composing or calculating the Bloomberg Barclays Indices. Neither Bloomberg nor Barclays is responsible for and has not participated in the determination of the timing of, prices at, or quantities of the products to be issued. Neither Bloomberg nor Barclays has any obligation or liability in connection with the administration, marketing or trading of the products.

The performance data shown represents past performance, which is not a guarantee of future results. Investment returns and principal value will fluctuate, so investors’ shares, when sold, may be worth more or less than their original cost. Current performance may be lower or higher than the performance data cited. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index. For performance data current to the most recent quarter-end, visit our website at www.vanguard.com/performance.

© 2019 The Vanguard Group, Inc. All rights reserved. Vanguard Marketing Corporation, Distributor of the Vanguard Funds. Advice offered through Vanguard Institutional Advisory Services (VIAS) are provided by Vanguard Advisors, Inc., a registered investment advisor.

Market and economic review

Economic and market overview

Global growth: Growth likely to shift down but not out.

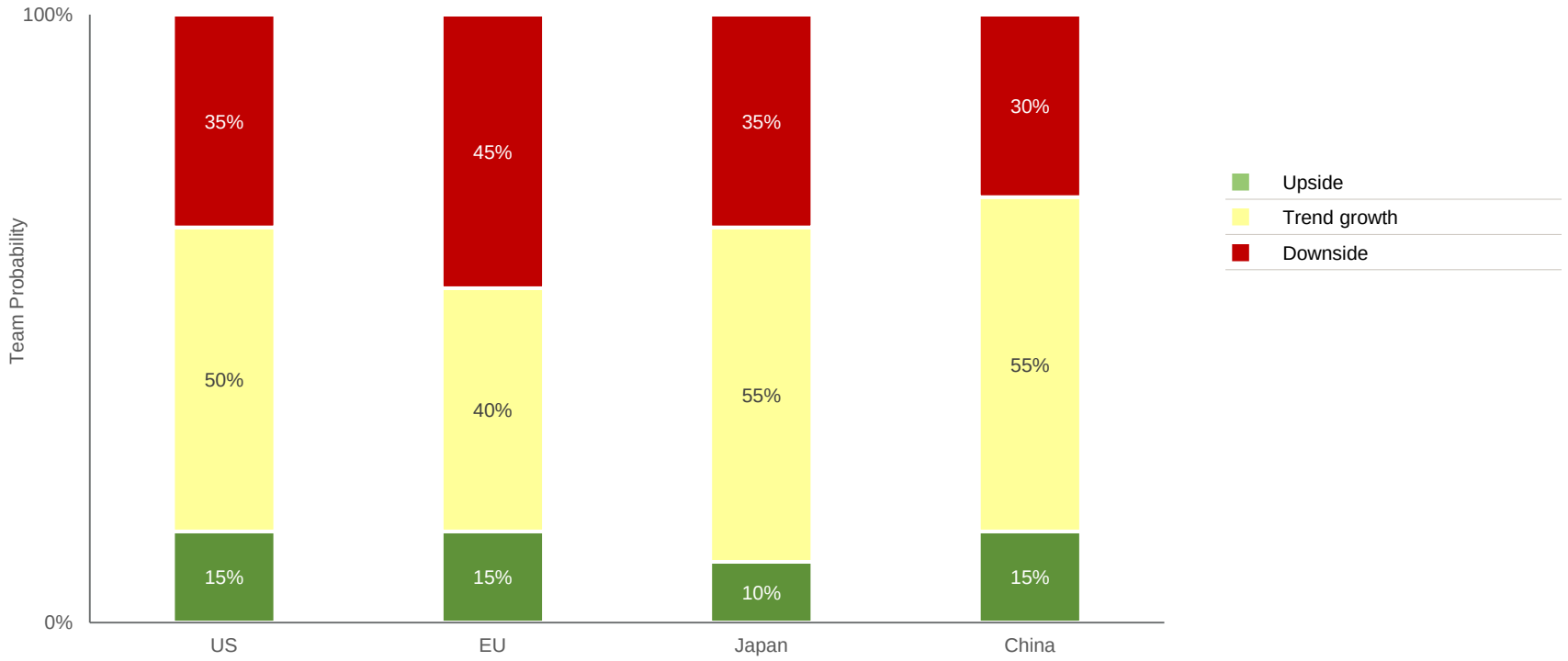
Inflation: Unlikely to accelerate as price expectations remain anchored.

Policy and interest rates: The tide of easy monetary policy is turning… very gradually.

Global asset returns: Expect higher risks and lower returns.

A probabilistic view of the growth outlook

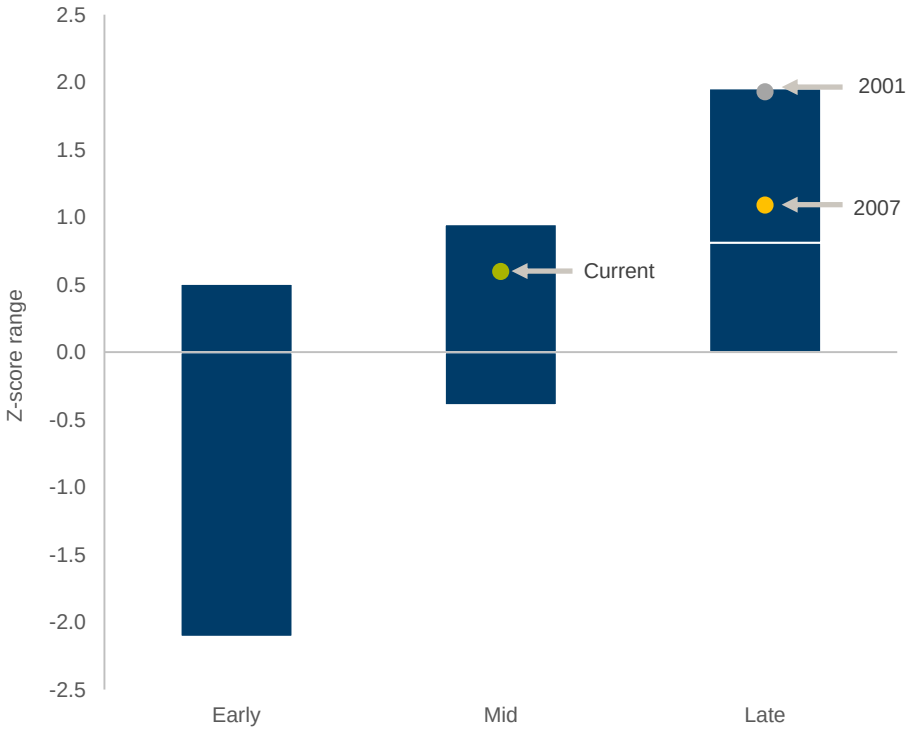
Risks skewed to the downside, expect a moderation to long-term trend



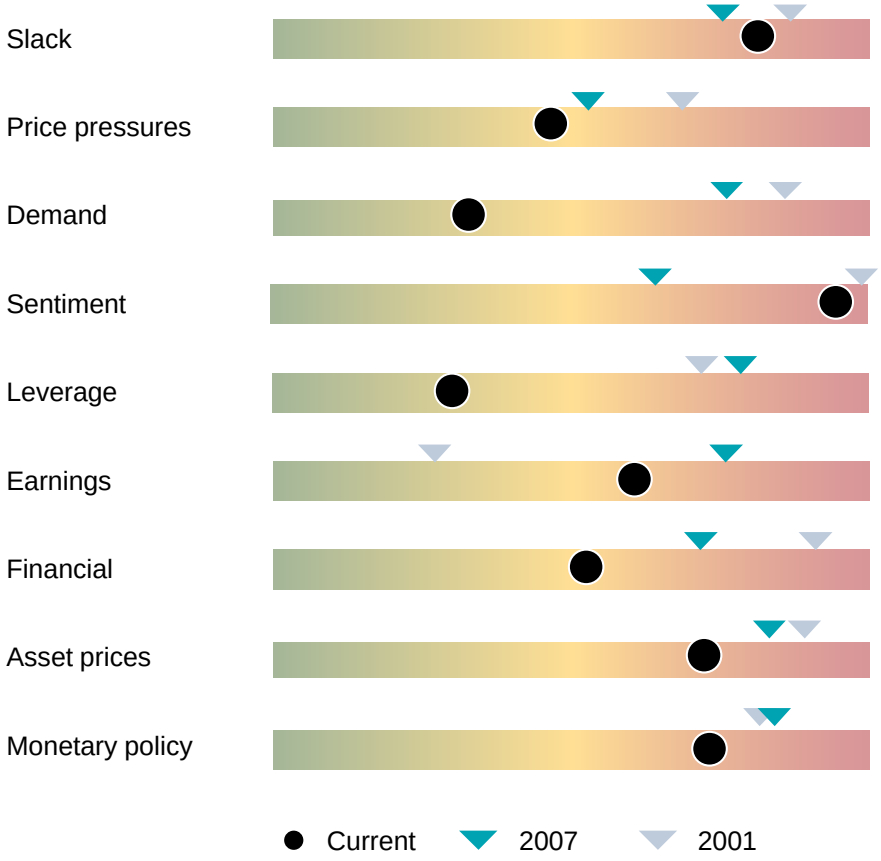
Source: Vanguard.

Still too early to call recession in the U.S. but risks are evident

Cyclical index just now moving toward later cycle levels



Lack of extremes broadly present across a range of indicators



Notes: Left figure displays the historical ranges of a cyclical index at various points in the business cycle. Index is shown as a z-score and weighted by first principal components of 25 economic indicators (below). The business cycle is determined by historical observations of the output gap. Right figure displays the underlying components of the cyclical index in right figure, presenting the current level relative to historical observations. The 2007 and 2001 data points indicate the index and component position 12 months prior to the onset of recession. Underlying indicators: slack = output gap, U3 and U6 unemployment rate gap relative to NAIRU. Price pressures = personal consumption expenditures (PCE), core PCE, average hourly earnings, unit labor costs. Demand = housing starts, residential investment, non-residential investment, durable goods consumption. Sentiment = business optimism, consumer sentiment, consumer confidence. Leverage = household financial obligations ratio, nonfinancial corporate debt, FRB Senior Loan Officer Opinion Survey for consumer and commercial and industrial credit terms. Earnings = corporate profits. Financial = Vanguard financial conditions index, yield curve (measured as the 10 year-3 month Treasury yield) Asset prices = Vanguard's fair-value CAPE, corporate OAS spread, high-yield OAS spread. Monetary policy = federal funds rate versus neutral rate estimated by the Laubach-Williams (2003) model. Data range is 1980 Q1-present.
Sources: Vanguard, Moody's Analytics Data Buffet, Federal Reserve Bank of St. Louis, Laubach-Williams (2003).

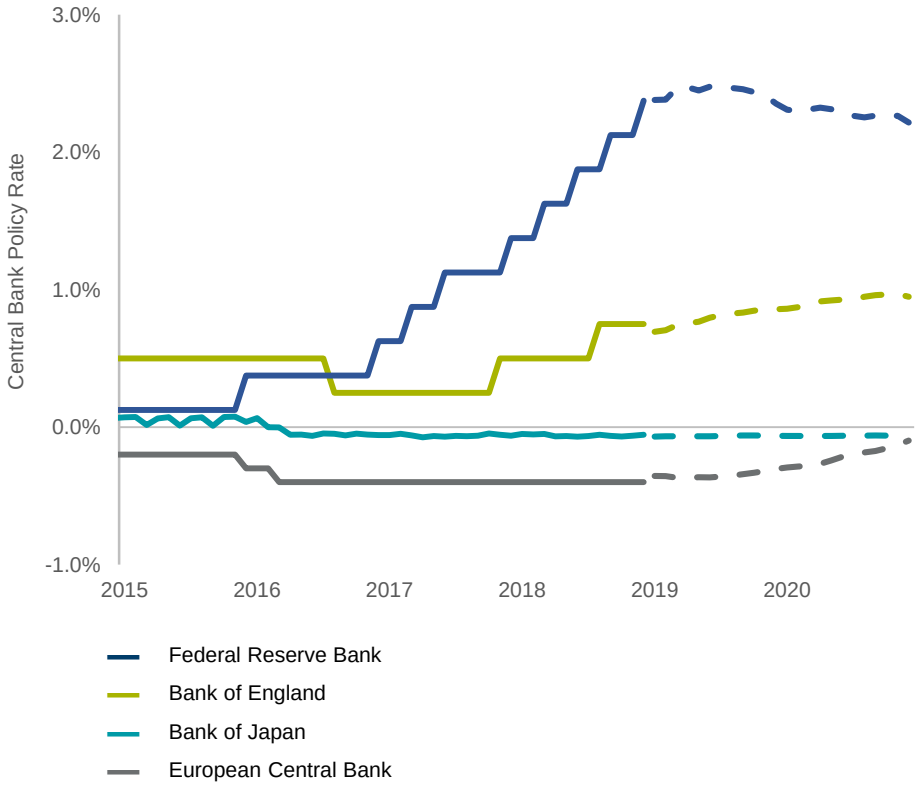
Global risks to the outlook

		Vanguard assessment of risks		
Risk	**Description**	**Negative scenario**	**Base case**	**Positive scenario**
Global monetary policy normalization	Extreme nature of existing policy stimulus, uncharted territory of quantitative tightening, and uncertainty about the neutral setting for policy (r*).	**25%** Policy mistake in the U.S.: Fed continues tightening beyond r* and the yield curve inverts. ECB/BOE hold off on normalization plans.	**60%** Soft landing in the U.S. at 2.75%–3%. Gradual ECB/BOE normalization commences. Global growth slowing back to trend.	**15%** Soft landing in the U.S. at 3% or higher. Global trend growth increases without global inflation.
Trade war and protectionism	Bilateral U.S.-China trade war continues to escalate. Tariffs can be increased further and non-tariff barriers implemented.	**20%** Trade war extends beyond tariffs with major retaliations from China. GDP growth impact could be more than 100 basis points.	**55%** Trade war escalates in intensity. Impact on the global economy of 30–50 basis points.	**25%** The U.S. and China reach a bilateral agreement. Tariffs are rolled back.
Instability of Chinese economy	Fears are rising about a hard landing in China, given the collateral damage of deleveraging and status of China-U.S. relations.	**30%** Chinese policymakers fail to provide enough stimulus. Headline growth falls below 6%.	**55%** Further policy easing will support domestic demand. Growth likely moderates to 6.0%–6.3% for 2019.	**15%** U.S.-China striking a trade deal and/or policy over-easing represent upside risks to growth.
Euro breakup risk	An escalation in tensions relating to Italy.	**10%** The Italian government maintains a loose fiscal policy resulting in eventual departure from the euro. This results in a wider crisis in the euro area.	**70%** The Italian government revises fiscal policy and tensions subside. Euro breakup concerns are lessened.	**20%** The Italian government backs down completely and submits a fiscal austerity plan and euro breakup concerns to subside.
Emerging-market debt crises	Key drivers of emerging-market cycles are global monetary policy divergence, the effect of the U.S. dollar on dollar denominated debt, and global/China demand for commodities.	**20%** Trade wars, a slowdown of the Chinese economy, or a strong U.S. dollar due to continued divergence of monetary policy lead to spillovers and broader EM crises.	**60%** Emerging-market debt crises remain contained to a few idiosyncratic cases.	**20%** U.S. dollar level normalizes as developed-market central banks commence normalization. Risk-on environment helps emerging markets undergo V-shape bounce-back.

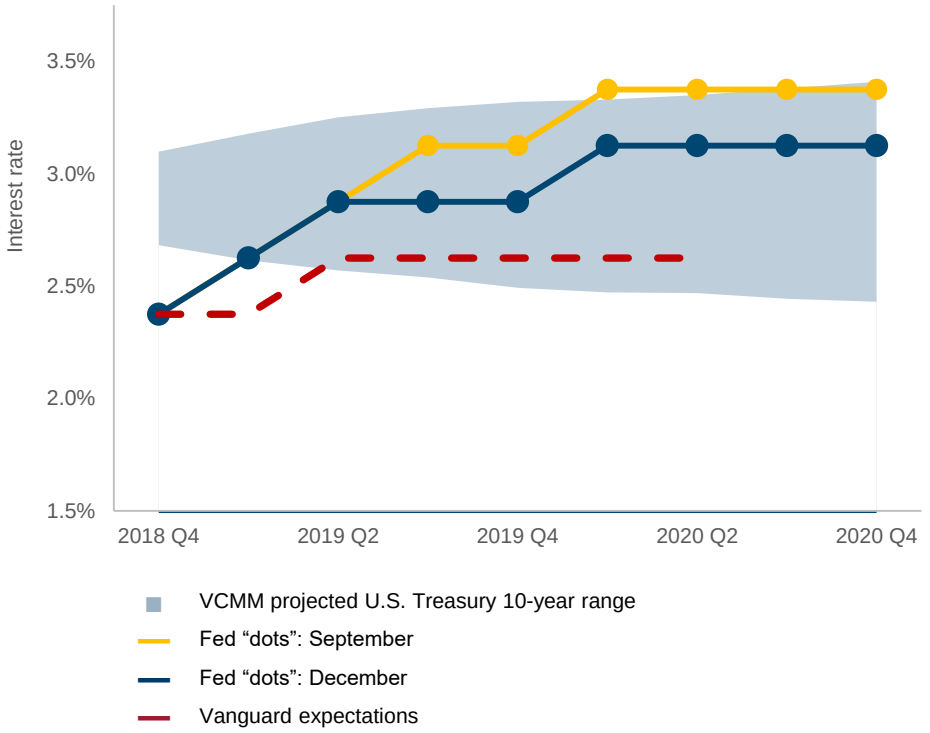
Note: Odds for each scenario are based on median responses to a poll of Vanguard's Global Economics and Capital Markets Outlook Team.
Source: Vanguard.

Monetary policy mistake is less likely as tightening path is drawn out

Market does not expect rate hikes in 2019



Resulting in lower risk of yield curve inversion



IMPORTANT: The projections and other information generated by the VCMM regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results, and are not guarantees of future results. Distribution of return outcomes from VCMM are derived from 10,000 simulations for each modeled asset class. Simulations as of September 30, 2018. Results from the model may vary with each use and over time. For more information, please see the important information slide.

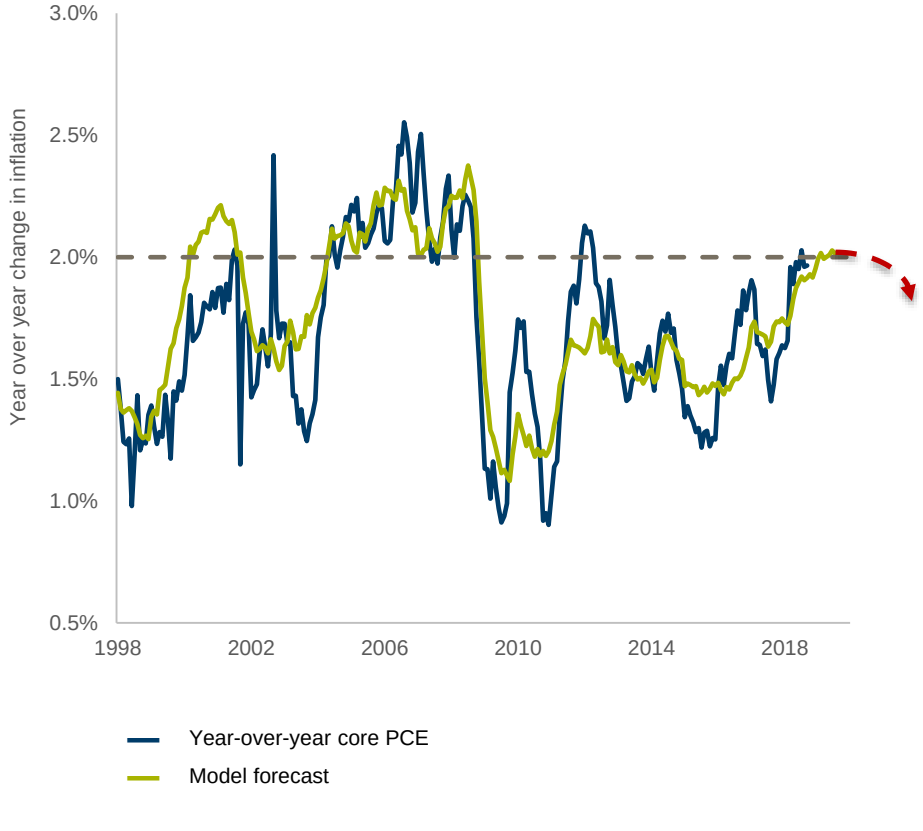
Sources: (LHS) Vanguard, using data from Bloomberg.
Notes: The solid lines represent Central Bank policy rates as defined as the Federal Funds Target Rate, Bank of Japan Unsecured Overnight Call Rate, UK Bank of England Official Bank Rate, Euro Overnight Index Average. Dotted lines represent expected future rates calculated via the futures market. Data as of December 31st, 2018. (RHS) Vanguard calculations, based on data from Thomson Reuters Datastream and Moody's Analytics Data Buffet; Federal Reserve Bank of New York. (RHS) FFR refers to federal funds rate. The U.S. 10-year Treasury path range uses the 35th and 65th percentile of projected Vanguard Capital Markets Model® (VCMM) path observations. VCMM is a proprietary financial simulation engine designed to help clients make effective asset allocation decisions.

A material rise in wage growth or inflation is not expected

Absent a productivity spike, higher wage growth unlikely



Runaway inflation remains unlikely

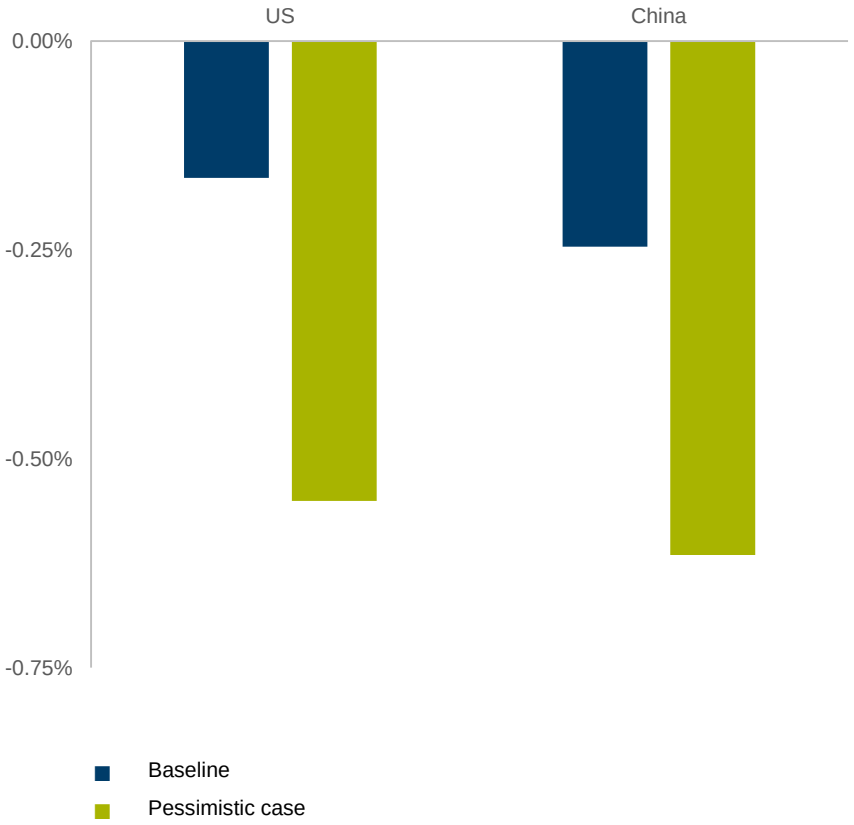


Notes: (LHS) Real wage growth is calculated as the growth rate of hourly wages as reported in the Employment Cost Index (ECI) minus core PCE inflation rate for that year. Trend for real wage growth is estimated as a centered three-year moving average of real wage growth. (RHS) Core PCE model is a root mean square error (RMSE)-weighted average of two models: a bottom-up model where we model the deviation of augmented Phillips curve fitted values to each major component in the core PCE and a top-down macro model. The RMSE is 0.35 for the bottom-up model and 0.24 for the top-down model. This leads to a 40% weight for the bottom-up model and a 60% weight for the top-down model in the weighted model.
Sources: (LHS) Congressional Budget Office, Bureau of Labor Statistics. (RHS) Vanguard calculations, based on Thomson Reuters Datastream, Bureau of Economic Analysis, Bureau of Labor Statistics, Philadelphia Federal Reserve Bank Survey of Professional Forecasters, Congressional Budget Office, and Bloomberg Commodity Index.

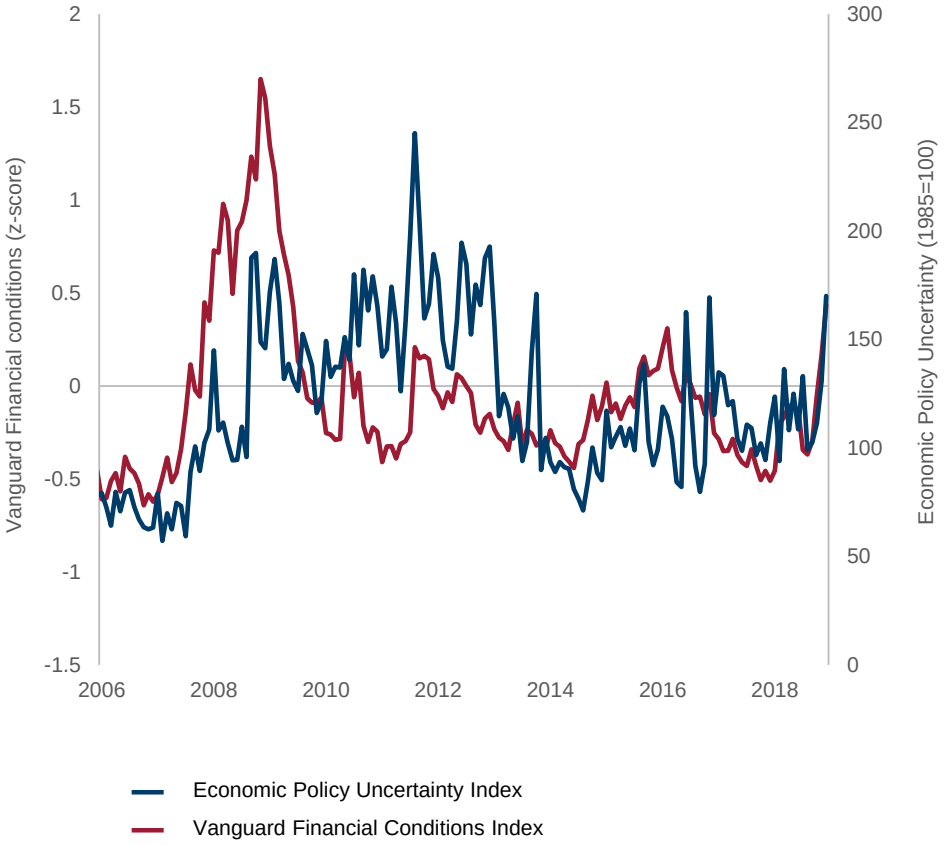
For institutional use only. Not for distribution to retail investors.

Trade, policy, and financial market uncertainty may prompt growth scales

Trade war impact on 2019 GDP



Twin shocks: Financial conditions and policy uncertainty

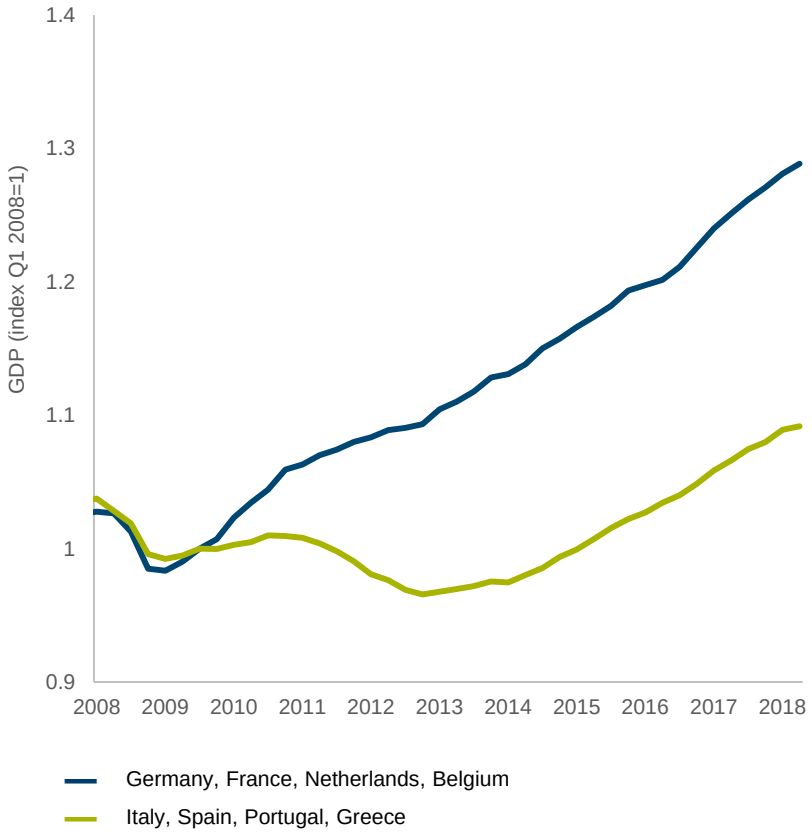


Notes: (LHS) Estimates for the impact on US GDP are based on simulations from the FRB-US model. The base case assumes 3% import and export price increases and +25 bps to the BBB spread, while the pessimistic case assumes 10% import and export prices and +100 bps to the BBB spread. Estimates for the impact on Chinese GDP are calculated using data on trade statistics, trade in value added, and tariff rates. (RHS) The Vanguard Financial Conditions is a dynamically-weighted index of 12 financial variables designed to capture overall financial conditions.

Sources: (LHS) Vanguard calculations, based on the Federal Reserve's FRB/US Model. (RHS) Vanguard calculations, based on data from Moody's Analytics Data Buffet, St. Louis Federal Reserve database, Thomson Reuters Datastream and Baker, Scott, Nicholas Bloom and Steven J. Davis, www.policyuncertainty.com .

Euro breakup flare-ups are likely if divergence continues

A tale of two recoveries



Italy's debt isn't the most worrying

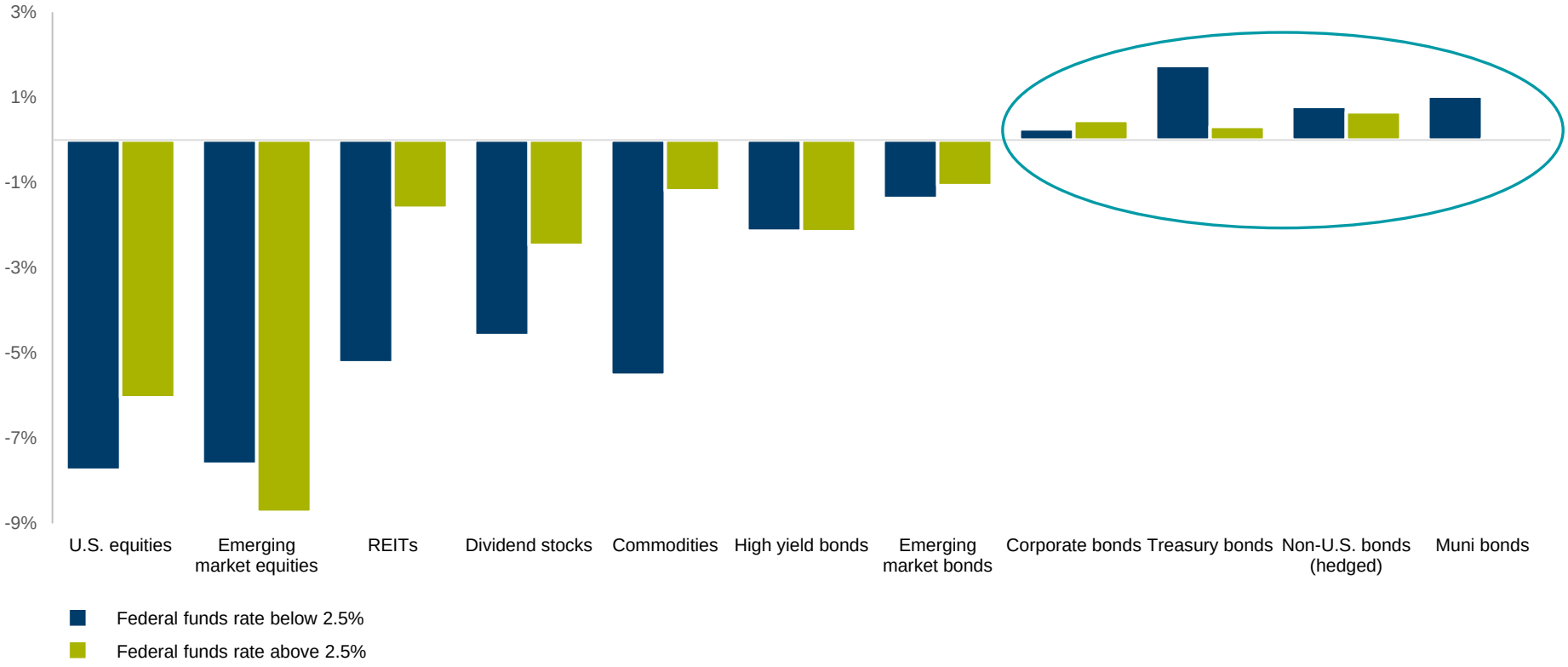


Sources: Vanguard calculations, based on Thomson Reuters Datastream and Bank of International Settlements

For institutional use only. Not for distribution to retail investors.

Bonds provide ballast during equity bear markets regardless of interest rate environment

Median return of various asset classes during the worst decile of monthly equity returns 1988-2018



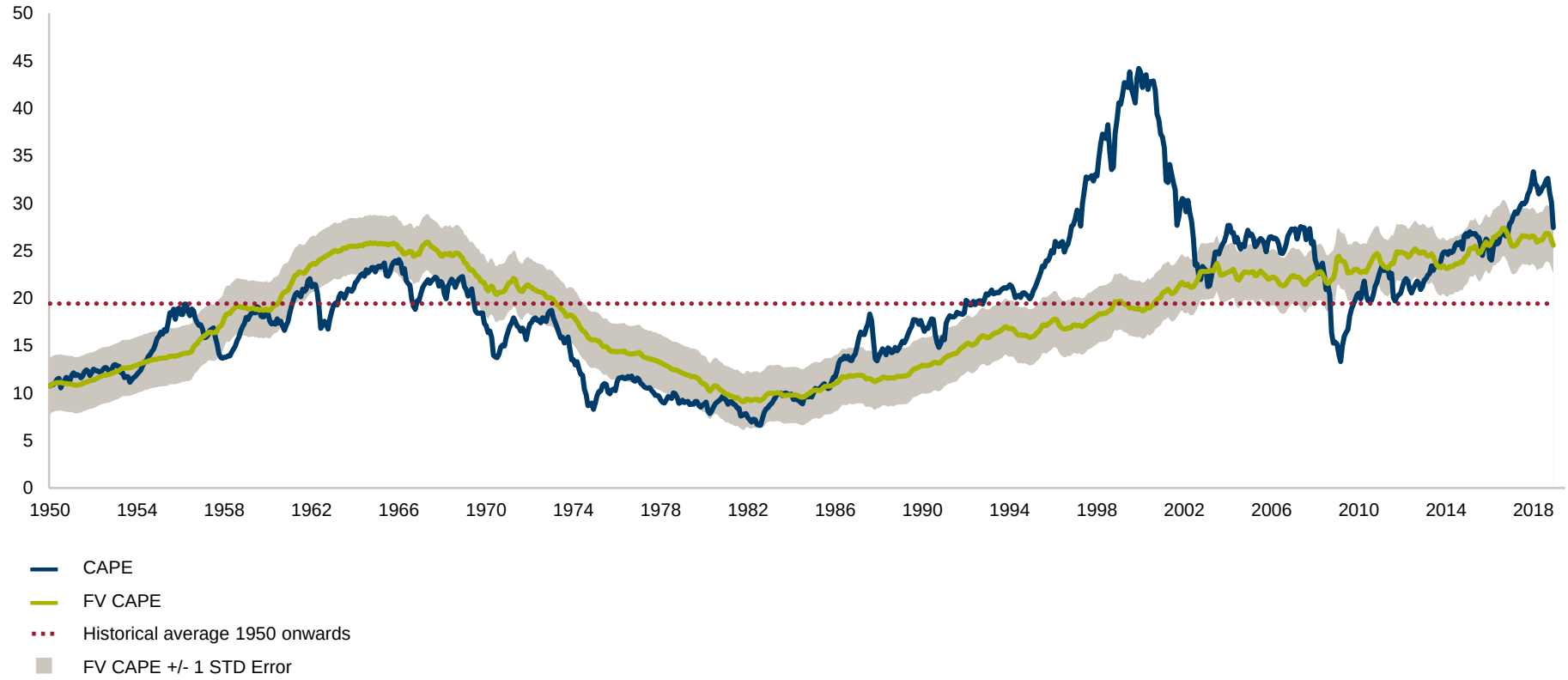
Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: Vanguard calculations based on data from Thomson Reuters Datastream, Bloomberg Barclays, HFRI, MSCI, FTSE, CRSP, S&P, and Dow Jones.
Notes: U.S. stocks represented by Dow Jones U.S. Total Stock Market Index through April 2005, MSCI US Broad Market Index through June 2013 and CRSP US Total Market Index thereafter; emerging markets stocks are represented by MSCI Emerging Markets Index; REITs by FTSE NAREIT Equity REIT Index; dividend stocks by Dow Jones U.S. Select Dividend Index; commodities by S&P GSCI Commodity Index; high yield bonds by Bloomberg Barclays U.S. Corporate High Yield Bond Index; emerging markets bonds by Bloomberg Barclays EM USD Aggregate Index; investment-grade corporate bonds by Bloomberg Barclays U.S. Corporate Index; U.S. Treasury bonds by Bloomberg Barclays U.S. Treasury Bond Index; U.S. Municipal Bonds by Bloomberg Barclays Municipal Bond Index; Hedge fund index by HFRI fund-weighted total return Index and international bonds by Bloomberg Barclays Global Aggregate ex-USD Bond Index. The Dow Jones U.S. Select Dividend Index starts in January 1992; Bloomberg Barclays EM USD Aggregate Index starts in January 1993; hedge fund data start in 1994 and Bloomberg Barclays Global Aggregate ex USD Bond Index starts in January 1990. All data provided through December 31, 2018.

U.S.: Equity valuations retreat to fair-value range

Vanguard's proprietary fair-value CAPE

Fair Value CAPE - US S&P 500

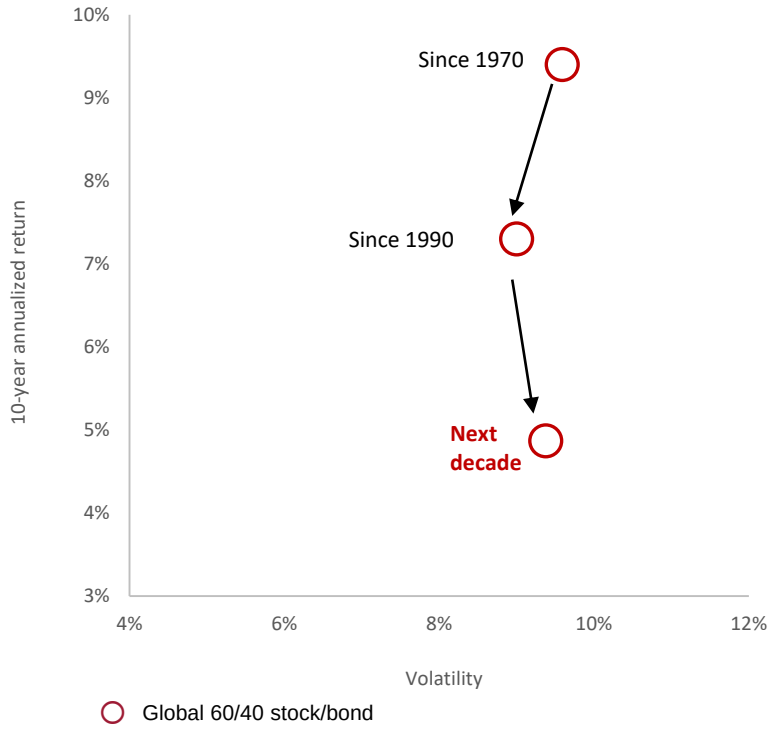


Note: "Fair-value CAPE" is based on a statistical model that corrects CAPE measures for the level of inflation expectations and for lower interest rates. The statistical model specification is a three-variable vector error correction (VEC), including equity-earnings yields, ten-year trailing inflation, and ten-year U.S. Treasury yields estimated over the period January 1940–December 2018. For details, see Vanguard's economic and investment outlook (Davis, Aliaga-Diaz, Westaway, Wang, Patterson, and Ahluwalia 2016).

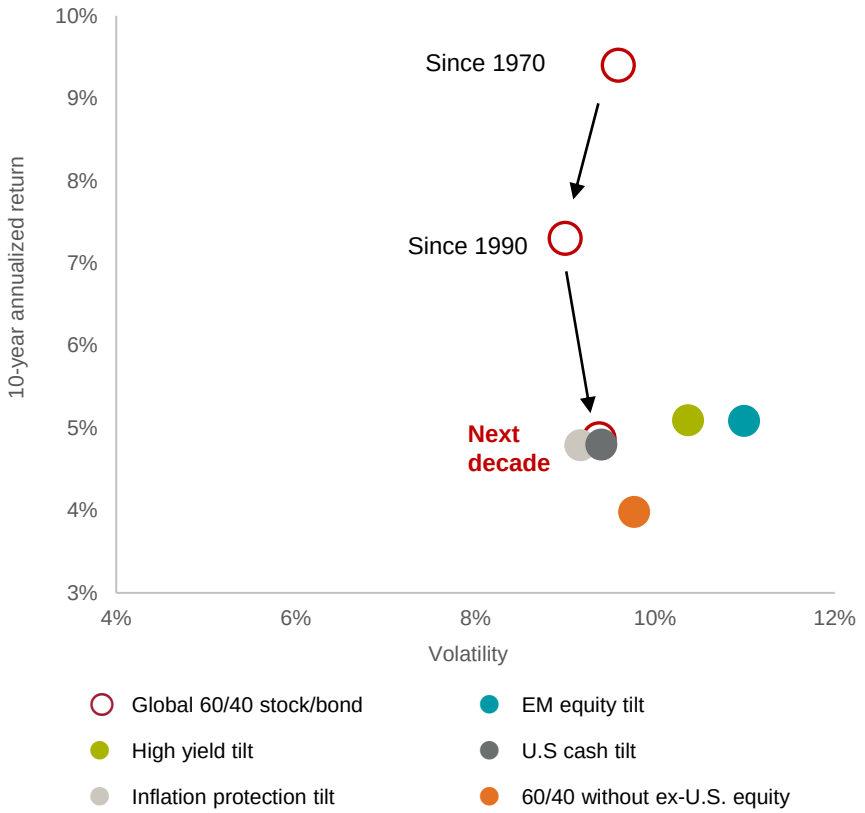
Sources: Vanguard calculations, based on Robert Shiller's website, at aida.wss.yale.edu/~shiller/data.htm., U.S. Bureau of Labor Statistics, and Federal Reserve Board.

Asset allocation for a challenging decade

A lower return orbit…



…That popular "active tilts" will likely fail to escape

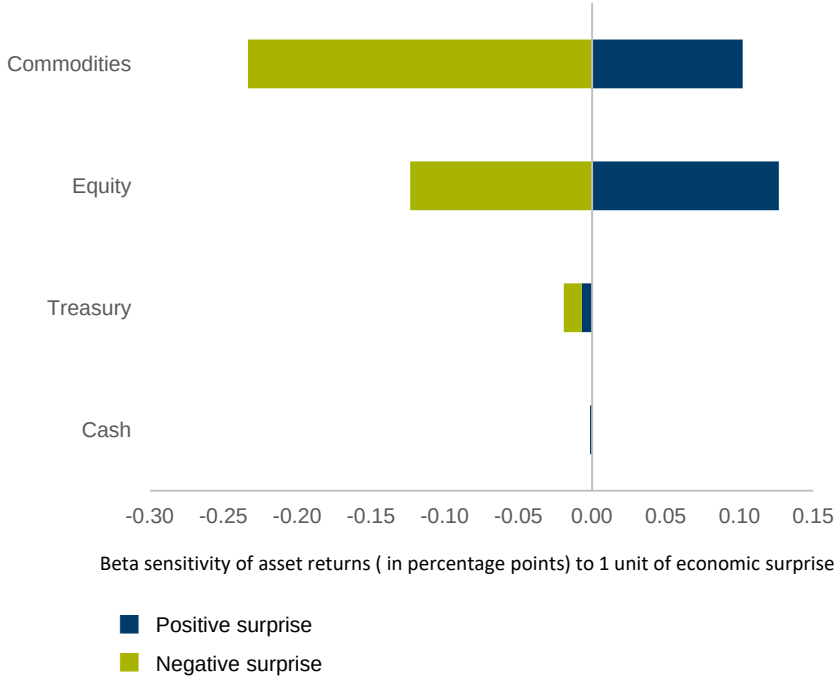


IMPORTANT: The projections and other information generated by the VCMM regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results, and are not guarantees of future results. Distribution of return outcomes from VCMM are derived from 10,000 simulations for each modeled asset class. Simulations as of September 30, 2018. Results from the model may vary with each use and over time. For more information, please see the important information slide.

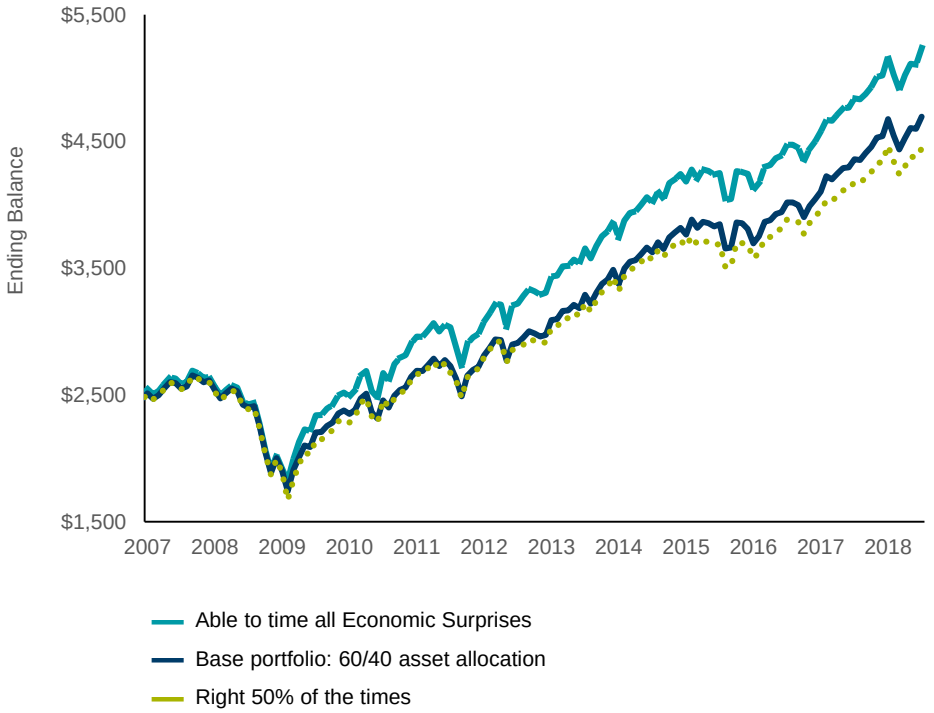
Notes: Summary statistics of 10,000 VCMM simulations for projected ten-year annualized nominal returns are as of September 2018 in U.S. dollars before costs. Historical returns are computed using indexes defined in "Indexes used in our historical calculations" on page 5 of Vanguard economic and market outlook for 2019: Down but not out. The global equity portfolio is 60% U.S. equity and 40% global ex-U.S. equity. The global bond portfolio is 70% U.S. bonds and 30% global ex-U.S. bonds. Portfolios with tilts include a 20% tilt to the asset specified funded from the fixed income allocation for the fixed income tilt and the equity allocation for the equity tilt.
Sources: Vanguard.

The impact of economic surprises on asset returns

High and low-risk assets respond differently to economic surprises



Very hard to capitalize on economic surprises



Notes: (LHS) The figure displays the results of regressing asset return for each category on the economic surprise index. The index is an equal-weighted index of surprises in the following macroeconomic data series: GDP final release, GDP preliminary release, ISM Manufacturing Index, retail sales, and nonfarm employment. The bars represent the change in asset returns in response to a 1-unit change in the economic surprise index. (RHS) The scenarios are based on the MSCI USA Index and the Bloomberg Barclays U.S. Aggregate Bond Index. • We start with a $1,000 investment in a base portfolio of 60% U.S. equities and 40% U.S. bonds. In advance of a positive economic surprise, we allocate 80% of the portfolio to equities and 20% to bonds. In advance of a negative economic surprise, we allocate 40% of the portfolio to equities and 60% to bonds.
Sources: Vanguard calculations, based on data from the U.S. Bureau of Labor Statistics, Bloomberg, Thomson Reuters, the BEA, and the ISM.

Fund summaries

Vanguard®

Vanguard Total Stock Market Index Fund (VITSX)

Equity characteristics

	VITSX	Benchmark
Number of stocks	3,508	3,607
Median market cap	$66.8B	$66.8B
P/E ratio (trailing earnings)	16.1x	16.1x
P/B ratio	2.7x	2.7x
Return on equity (5-year average)	15.0%	15.0%
Earnings growth rate (5-year)	8.1%	8.1%
Equity yield (dividend)	2.0%	2.0%
Foreign holdings	0.1%	—
Turnover (fiscal year end)	3.4%	—
Short-term reserves	0.0%	N/A

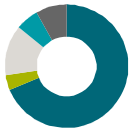
Sector diversification as a % of common stock



Sector	VITSX	Benchmark
Technology	19.5%	19.4%
Financials	19.4	19.5
Health Care	13.9	13.9
Consumer Services	13.6	13.6
Industrials	12.9	12.9
Consumer Goods	7.9	7.9
Oil & Gas	5.0	5.0
Utilities	3.3	3.3
Basic Materials	2.5	2.5
Telecommunications	2.0	2.0

Sector categories are based on the Industry Classification Benchmark ("ICB"), except for the "Other" category (if applicable), which includes securities that have not been provided an ICB classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VITSX	Benchmark
More than $26	68.6%	68.4%
$20 to $26	3.9	3.9
$8 to $20	13.3	13.3
$4 to $8	5.9	5.9
Below $4	8.2	8.5

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VITSX	N/A	N/A	-0.00	11.35	0.71
Primary benchmark	1.00	1.00	N/A	11.35	0.71
Broad-based benchmark	1.00	1.00	N/A	11.37	0.70

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Dow Jones U.S. Total Stock Market Float Adjusted Index.

Ten largest holdings

	% of total net assets
Microsoft Corp.	3.1%
Apple Inc.	2.7
Amazon.com Inc.	2.5
Alphabet Inc.	2.5
Berkshire Hathaway Inc.	1.6
Johnson & Johnson	1.4
JPMorgan Chase & Co.	1.3
Facebook Inc.	1.3
Exxon Mobil Corp.	1.1
Pfizer Inc.	1.0
Top ten as a % of total net assets	**18.5%**

The holdings listed exclude any temporary cash investments and equity index products.

FOR FINANCIAL ADVISORS AND INSTITUTIONS ONLY. NOT FOR PUBLIC DISTRIBUTION.

Vanguard®

Vanguard Total International Stock Index Fund (VTSNX)

Equity characteristics

	VTSNX	Benchmark
Number of stocks	6,374	6,001
Median market cap	$23.7B	$23.7B
P/E ratio (trailing earnings)	12.0x	12.1x
P/B ratio	1.4x	1.4x
Return on equity (5-year average)	12.4%	12.4%
Earnings growth rate (5-year)	9.1%	9.1%
Equity yield (dividend)	3.3%	3.3%
Turnover (fiscal year end)	3.4%	—
Short-term reserves	0.0%	N/A

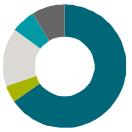
Sector diversification as a % of common stock



Sector	VTSNX	Benchmark
Financials	25.0%	25.0%
Consumer Goods	14.4	14.5
Industrials	14.1	14.1
Consumer Services	9.2	9.2
Health Care	8.4	8.3
Technology	8.0	8.0
Basic Materials	7.1	7.1
Oil & Gas	6.8	6.8
Telecommunications	3.7	3.7
Utilities	3.3	3.3

Sector categories are based on the Industry Classification Benchmark ("ICB"), except for the "Other" category (if applicable), which includes securities that have not been provided an ICB classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VTSNX	Benchmark
More than $12	65.4%	65.4%
$9 to $12	4.4	4.4
$4 to $9	15.2	15.1
$2 to $4	6.2	6.2
Below $2	8.8	8.9

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VTSNX	N/A	N/A	0.02	11.34	0.32
Primary benchmark	0.98	0.97	N/A	11.60	0.30
Broad-based benchmark	0.98	0.97	N/A	11.60	0.30

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: FTSE Global All Cap ex US Index.

Market allocation

	% of common stock	
	VTSNX	Benchmark
Japan	17.4%	17.7%
United Kingdom	11.9	11.7
China	6.8	6.8
France	6.5	6.5
Canada	6.3	6.5
Germany	5.7	5.7
Switzerland	5.5	5.5
Australia	4.8	4.8
Korea	3.5	3.5
Taiwan	3.2	3.2
Other	28.4	28.1

Ten largest holdings

	% of total net assets
Royal Dutch Shell plc	1.1%
Nestle SA	1.1
Tencent Holdings Ltd.	1.1
Samsung Electronics Co. Ltd.	0.9
Novartis AG	0.9
Roche Holding AG	0.8
Taiwan Semiconductor Manufacturing Co. Ltd.	0.8
HSBC Holdings plc	0.8
Alibaba Group Holding Ltd.	0.8
Toyota Motor Corp.	0.7
Top ten as a % of total net assets	**9.0%**

The holdings listed exclude any temporary cash investments and equity index products.

FOR FINANCIAL ADVISORS AND INSTITUTIONS ONLY. NOT FOR PUBLIC DISTRIBUTION.

Vanguard Total Bond Market Index Fund (VBTIX)

Fixed income characteristics

	VBTIX	Benchmark
Number of bonds	8,582	10,248
Average duration	5.9 years	5.9 years
Average effective maturity	8.3 years	8.2 years
Turnover (fiscal year end)	60.0%	—
Short-term reserves	0.0%	N/A

Distribution by issuer type

	% of total net assets
Treasury/Agency	43.3%
Government Mortgage-Backed	22.1%
Industrial	16.0%
Finance	8.7%
Foreign	5.2%
Commercial Mortgage-Backed	1.9%
Utilities	1.9%
Other	0.5%
Asset-Backed	0.4%

Hedged non-U.S. dollar-denominated bonds are included in the sector of issuer, not as part of the foreign category.

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VBTIX	N/A	N/A	-0.00	2.97	0.36
Primary benchmark	0.99	1.01	N/A	2.93	0.38
Broad-based benchmark	0.99	1.01	N/A	2.93	0.38

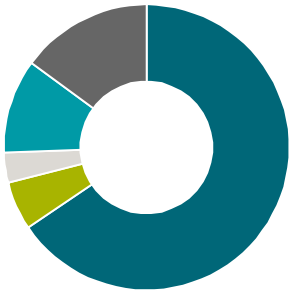
R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Spliced Bloomberg Barclays U.S. Aggregate Float Adjusted Index.

Distribution by effective maturity as a % of funds



Under 1 Year	0.4%
1 - 3 Years	23.8
3 - 5 Years	19.4
5 - 10 Years	39.5
10 - 20 Years	3.7
20 - 30 Years	12.7
Over 30 Years	0.5

Distribution by credit quality as a % of funds



U.S. Government	65.5%
Aaa	5.5
Aa	3.4
A	10.7
Baa	14.9
Less Than Baa	0.0

Credit-quality ratings are measured on a scale that generally ranges from AAA (highest) to D (lowest). "NR" is used to classify securities for which a rating is not available. NR securities may include a fund's investment in Vanguard Market Liquidity Fund or Vanguard Municipal Cash Management Fund, each of which invests in high-quality money market instruments and may serve as a cash management vehicle for the Vanguard funds, trusts, and accounts. U.S. Treasury, U.S. Agency, and U.S. Agency mortgage-backed securities appear under "U.S. Government." Credit-quality ratings for each issue are obtained from Barclays using ratings derived from Moody's Investors Service (Moody's), Fitch Ratings (Fitch), and Standard & Poor's (S&P). When ratings from all three agencies are available, the median rating is used. When ratings are available from two of the agencies, the lower rating is used. When one rating is available, that rating is used.

FOR FINANCIAL ADVISORS AND INSTITUTIONS ONLY. NOT FOR PUBLIC DISTRIBUTION.

Vanguard Real Estate Index Fund (VGSNX)

Equity characteristics

	VGSNX	Benchmark
Number of stocks	189	193
Median market cap	$13.1B	$12.3B
P/E ratio (trailing earnings)	30.0x	30.0x
P/B ratio	2.2x	2.2x
Return on equity (5-year average)	6.1%	6.1%
Earnings growth rate (5-year)	13.8%	14.0%
Equity yield (dividend)	4.2%	4.2%
Foreign holdings	0.0%	—
Turnover (fiscal year end)	6.1%	—
Short-term reserves	0.0%	N/A

Subindustry diversification as a % of common stock



Sector	VGSNX	Benchmark
Specialized REITs	32.0%	32.0%
Retail REITs	14.6	14.6
Residential REITs	14.1	14.0
Health Care REITs	9.8	9.9
Office REITs	9.8	9.9
Industrial REITs	6.8	6.7
Hotel & Resort REITs	4.9	4.9
Diversified REITs	4.6	4.6
Real Estate Services	2.5	2.5
Real Estate Development	0.4	0.4
Real Estate Operating Companies	0.3	0.3

Sector categories are based on the Global Industry Classification Standard ("GICS"), except for the "Other" category (if applicable), which includes securities that have not been provided a GICS classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VGSNX	Benchmark
More than $26	30.1%	29.3%
$20 to $26	9.2	9.2
$8 to $20	25.0	25.1
$4 to $8	16.7	17.1
Below $4	19.0	19.4

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VGSNX	N/A	N/A	-0.01	13.49	0.10
Primary benchmark	1.00	1.00	N/A	13.47	0.11
Broad-based benchmark	0.36	0.71	N/A	11.37	0.70

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Dow Jones U.S. Total Stock Market Float Adjusted Index.

Ten largest holdings

	% of total net assets
Vanguard Real Estate II Index Fund	11.3%
American Tower Corp.	6.3
Simon Property Group Inc.	4.7
Crown Castle International Corp.	4.0
Prologis Inc.	3.3
Public Storage	2.9
Equinix Inc.	2.5
Welltower Inc.	2.3
Equity Residential	2.2
AvalonBay Communities Inc.	2.2
Top ten as a % of total net assets	**41.7%**

The holdings listed exclude any temporary cash investments and equity index products.

FOR FINANCIAL ADVISORS AND INSTITUTIONS ONLY. NOT FOR PUBLIC DISTRIBUTION.

Important information

IMPORTANT: The projections or other information generated by the Vanguard Capital Markets Model® (VCMM) regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results, and are not guarantees of future results. VCMM results will vary with each use and over time.

The VCMM projections are based on a statistical analysis of historical data. Future returns may behave differently from the historical patterns captured in the VCMM. More importantly, the VCMM may be underestimating extreme negative scenarios unobserved in the historical period on which the model estimation is based.

The Vanguard Capital Markets Model® is a proprietary financial simulation tool developed and maintained by Vanguard's primary investment research and advice teams. The model forecasts distributions of future returns for a wide array of broad asset classes. Those asset classes include U.S. and international equity markets, several maturities of the U.S. Treasury and corporate fixed income markets, international fixed income markets, U.S. money markets, commodities, and certain alternative investment strategies. The theoretical and empirical foundation for the Vanguard Capital Markets Model is that the returns of various asset classes reflect the compensation investors require for bearing different types of systematic risk (beta). At the core of the model are estimates of the dynamic statistical relationship between risk factors and asset returns, obtained from statistical analysis based on available monthly financial and economic data from as early as 1960. Using a system of estimated equations, the model then applies a Monte Carlo simulation method to project the estimated interrelationships among risk factors and asset classes as well as uncertainty and randomness over time. The model generates a large set of simulated outcomes for each asset class over several time horizons. Forecasts are obtained by computing measures of central tendency in these simulations. Results produced by the tool will vary with each use and over time.

The Vanguard Lifecycle Model (VLCM) is designed to identify the product design that represents the best investment solution for a theoretical, representative investor who uses the target-date funds to accumulate wealth for retirement. The VLCM generates an optimal custom glide path for a participant population by assessing the trade-offs between the expected (median) wealth accumulation and the uncertainty about that wealth outcome, for thousands of potential glide paths. The VLCM does this by combining two set of inputs: the asset class return projections from the VCMM and the average characteristics of the participant population. Along with the optimal custom glide path, the VLCM generates a wide range of portfolio metrics such as a distribution of potential wealth accumulation outcomes, risk and return distributions for the asset allocation, and probability of ruin, such as the odds of participants depleting their wealth by age 95.

The VLCM inherits the distributional forecasting framework of the VCMM and applies to it the calculation of wealth outcomes from any given portfolio.

The most impactful drivers of glide path changes within the VLCM tend to be risk aversion, the presence of a defined benefit plan, retirement age, savings rate and starting compensation. The VLCM chooses among glide paths by scoring them according to the utility function described and choosing the one with the highest score. The VLCM does not optimize the levels of spending and contribution rates. Rather, the VLCM optimizes the glide path for a given customizable level of spending, growth rate of contributions and other plan sponsor characteristics.

A full dynamic stochastic life-cycle model, including optimization of a savings strategy and dynamic spending in retirement is beyond the scope of this framework.

Important information

For more information about any fund, visit vanguard.com or call 866-499-8473 to obtain a prospectus or, if available, a summary prospectus. Investment objectives, risks, charges, expenses, and other important information are contained in the prospectus; read and consider it carefully before investing.

Vanguard ETF® Shares are not redeemable with the issuing Fund other than in very large aggregations worth millions of dollars. Instead, investors must buy or sell Vanguard ETF Shares in the secondary market and hold those shares in a brokerage account. In doing so, the investor may incur brokerage commissions and may pay more than net asset value when buying and receive less than net asset value when selling.

Mutual funds and all investments are subject to risk, including the possible loss of the money you invest. Prices of mid- and small-cap stocks often fluctuate more than those of large-company stocks. Investments in stocks or bonds issued by non-U.S. companies are subject to risks including country/regional risk and currency risk. These risks are especially high in emerging markets. Funds that concentrate on a relatively narrow sector face the risk of higher share-price volatility. It is possible that tax-managed funds will not meet their objective of being tax-efficient. Because company stock funds concentrate on a single stock they are considered riskier than diversified stock funds.

Investments in bond funds are subject to the risk that an issuer will fail to make payments on time, and that bond prices will decline because of rising interest rates or negative perceptions of an issuer's ability to make payments. High-yield bonds generally have medium- and lower-range credit quality ratings and are therefore subject to a higher level of credit risk than bonds with higher credit quality ratings. Although the income from a municipal bond fund is exempt from federal tax, you may owe taxes on any capital gains realized through the fund's trading or through your own redemption of shares. For some investors, a portion of the fund's income may be subject to state and local taxes, as well as to the federal Alternative Minimum Tax. Diversification does not ensure a profit or protect against a loss.

Investments in Target Retirement Funds or Trusts are subject to the risks of their underlying funds. The year in the fund name refers to the approximate year (the target date) when an investor in the fund or trust would retire and leave the workforce. The fund or trust will gradually shift its emphasis from more aggressive investments to more conservative ones based on its target date. An investment in the Target Retirement Fund or Trust is not guaranteed at any time, including on or after the target date.

Vanguard collective trusts are not mutual funds. They are collective trusts available only to tax-qualified plans and their eligible participants. Investment objectives, risks, charges, expenses, and other important information should be considered carefully before investing. The collective trust mandates are managed by Vanguard Fiduciary Trust Company, a wholly owned subsidiary of The Vanguard Group, Inc.

A stable value investment is neither insured nor guaranteed by the U.S. government. There is no assurance that the investment will be able to maintain a stable net asset value, and it is possible to lose money in such an investment.

Factor funds are subject to investment style risk, which is the chance that returns from the types of stocks in which the fund invests will trail returns from U.S. stock markets. Factor funds are subject to manager risk, which is the chance that poor security selection will cause the fund to underperform relevant benchmarks or other funds with a similar investment objective.

The information contained herein does not constitute tax advice, and cannot be used by any person to avoid tax penalties that may be imposed under the Internal Revenue Code. We recommend that you consult a tax or financial advisor about your individual situation.

Advice services are provided by Vanguard Advisers, Inc., a registered investment advisor.

Brokerage services are plan specific and may be provided by TD Ameritrade, Inc., member FINRA/SIPC or Vanguard Brokerage Services, a division of Vanguard Marketing Corporation, member FINRA/SIPC. Refer to Vanguard's plan documents for information on the applicable brokerage services provider. TD Ameritrade and Vanguard are separate and unaffiliated firms, and are not responsible for each other's services or policies. TD Ameritrade is a trademark jointly owned by TD Ameritrade IP Company, Inc., and the Toronto-Dominion Bank. Used with permission.

CFA® and Chartered Financial Analyst® are trademarks owned by CFA Institute. Financial Engines is a registered trademark and Financial Engines Investment Advisor service is a registered service mark of Financial Engines, Inc. Financial Engines Advisors L.L.C., a federally registered investment advisor and wholly owned subsidiary of Financial Engines, Inc., provides all advisory services. The Vanguard Group has partnered with Financial Engines to provide the Vanguard Managed Account Program and Personal Online Advisor, powered by Financial Engines.

CGS identifiers have been provided by CUSIP Global Services, managed on behalf of the American Bankers Association by Standard & Poor's Financial Services, LLC, and are not for use or dissemination in a manner that would serve as a substitute for any CUSIP service. The CUSIP Database, © 2019 American Bankers Association. "CUSIP" is a registered trademark of the American Bankers Association.

Vanguard Marketing Corporation, Distributor of the Vanguard Funds. U.S. Patent Nos. 6,879,964; 7,337,138; 7,720,749; 7,925,573; 8,090,646; and 8,417,623.

Important information

BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. BARCLAYS® is a trademark and service mark of Barclays Bank Plc, used under license. Bloomberg Finance L.P. and its affiliates, including Bloomberg Index Services Limited ("BISL") (collectively, "Bloomberg"), or Bloomberg's licensors own all proprietary rights in the Bloomberg Barclays Indices.

The products are not sponsored, endorsed, issued, sold or promoted by "Bloomberg or Barclays". Bloomberg and Barclays make no representation or warranty, express or implied, to the owners or purchasers of the products or any member of the public regarding the advisability of investing in securities generally or in the products particularly or the ability of the Bloomberg Barclays Indices to track general bond market performance. Neither Bloomberg nor Barclays has passed on the legality or suitability of the products with respect to any person or entity. Bloomberg's only relationship to Vanguard and the products are the licensing of the Bloomberg Barclays Indices which are determined, composed and calculated by BISL without regard to Vanguard or the products or any owners or purchasers of the products. Bloomberg has no obligation to take the needs of the products or the owners of the products into consideration in determining, composing or calculating the Bloomberg Barclays Indices. Neither Bloomberg nor Barclays is responsible for and has not participated in the determination of the timing of, prices at, or quantities of the products to be issued. Neither Bloomberg nor Barclays has any obligation or liability in connection with the administration, marketing or trading of the products.

London Stock Exchange Group companies include FTSE International Limited ("FTSE"), Frank Russell Company ("Russell"), MTS Next Limited ("MTS"), and FTSE TMX Global Debt Capital Markets Inc. ("FTSE TMX"). All rights reserved. "FTSE®", "Russell®", "MTS®", "FTSE TMX®" and "FTSE Russell" and other service marks and trademarks related to the FTSE or Russell Indexes are trademarks of the London Stock Exchange Group companies and are used by FTSE, MTS, FTSE TMX, and Russell under license. All information is provided for information purposes only. No responsibility or liability can be accepted by the London Stock Exchange Group companies nor its licensors for any errors or for any loss from use of this publication. Neither the London Stock Exchange Group companies nor any of its licensors make any claim, prediction, warranty or representation whatsoever, expressly or impliedly, either as to the results to be obtained from the use of the FTSE Indexes or the fitness or suitability of the Indexes for any particular purpose to which they might be put.

The index is a product of S&P Dow Jones Indices LLC ("SPDJI"), and has been licensed for use by Vanguard. Standard & Poor's® and S&P® are registered trademarks of Standard & Poor's Financial Services LLC ("S&P"); Dow Jones® is a registered trademark of Dow Jones Trademark Holdings LLC ("Dow Jones"); S&P® and S&P 500® are trademarks of S&P; and these trademarks have been licensed for use by SPDJI and sublicensed for certain purposes by Vanguard. Vanguard product(s) are not sponsored, endorsed, sold or promoted by SPDJI, Dow Jones, S&P, their respective affiliates and none of such parties make any representation regarding the advisability of investing in such product(s) nor do they have any liability for any errors, omissions, or interruptions of the index.

Morningstar data ©2019 Morningstar, Inc. All Rights Reserved. The information contained herein: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information. Past performance is no guarantee of future results.

The Russell Indexes and Russell® are registered trademarks of Russell Investments and have been licensed for use by The Vanguard Group. The products are not sponsored, endorsed, sold or promoted by Russell Investments and Russell Investments makes no representation regarding the advisability of investing in the products.

"Dividend Achievers" is a trademark of The NASDAQ OMX Group, Inc. (collectively, with its affiliates, "NASDAQ OMX") and has been licensed for use by The Vanguard Group, Inc. Vanguard mutual funds are not sponsored, endorsed, sold, or promoted by NASDAQ OMX and NASDAQ OMX makes no representation regarding the advisability of investing in the funds. NASDAQ OMX MAKES NO WARRANTIES AND BEARS NO LIABILITY WITH RESPECT TO THE VANGUARD MUTUAL FUNDS.

The funds or securities referred to herein are not sponsored, endorsed, or promoted by MSCI, and MSCI bears no liability with respect to any such funds or securities. The prospectus or the Statement of Additional Information contains a more detailed description of the limited relationship MSCI has with Vanguard and any related funds.

The Global Industry Classification Standard ("GICS") was developed by and is the exclusive property and a service mark of MSCI Inc. ("MSCI") and Standard and Poor's, a division of McGraw-Hill Companies, Inc. ("S&P") and is licensed for use by Vanguard. Neither MSCI, S&P nor any third party involved in making or compiling the GICS or any GICS classification makes any express or implied warranties or representations with respect to such standard or classification (or the results to be obtained by the use thereof), and all such parties hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any such standard or classification. Without limiting any of the foregoing, in no event shall MSCI, S&P, any of its affiliates or any third party involved in making or compiling the GICS or any GICS classification have any liability for any direct, indirect, special, punitive, consequential or any other damages (including lost profits) even if notified of the possibility of such damages.

Apple®, iPhone®, and iPad® are trademarks of Apple Inc., registered in the United States and other countries. App Store is a service mark of Apple Inc. Android™ is a trademark of Google Inc.

© 2019 The Vanguard Group, Inc. All rights reserved.

Rev_012019